

MAHIDOL UNIVERSITY

Faculty of Information and Communication Technology



Lecture 1

Introduction to Data Communication

August 2026

ปัญญาของแผ่นดิน Wisdom of the Land

มหาวิทยาลัยมหิดล
Mahidol University

ห้ามมิให้นักศึกษาทำซ้ำ ดัดแปลง หรือใช้ประโยชน์จากงานอันมีลิขสิทธิ์ปรากฏอยู่ในระบบการศึกษาอิเล็กทรอนิกส์ (E-Learning) ของมหาวิทยาลัย ไม่ว่าจะทั้งหมดหรือบางส่วน โดยไม่ได้รับอนุญาตจากมหาวิทยาลัย นอกเหนือจากการศึกษาส่วนบุคคล ทั้งนี้ การทำซ้ำ ดัดแปลง หรือเผยแพร่ต่อสาธารณชนซึ่งงานอันมีลิขสิทธิ์ **จะมีโทษปรับตั้งแต่ 20,000 บาท ถึง 200,000 บาท และหากเป็นการกระทำเพื่อการค้า จะมีโทษจำคุกตั้งแต่ 6 เดือน ถึง 10 ปี หรือปรับตั้งแต่ 100,000 บาท ถึง 800,000 บาท หรือทั้งจำทั้งปรับ**

Unless allowed by Mahidol University, the User shall not copy, modify, or exploit in part or in whole of the copyrighted materials on the Platform, other than for your own individual study. The copy, modify or communication to public of the copyrighted materials shall be inflicted with a fine from 20,000 Baht up to 200,000 Baht. If the offence is committed with the commercial purpose, the offender shall be inflicted with imprisonment for a term from 6 months up to 10 years or a fine from 100,000 Baht up to 800,000 Baht or both imprisonment and fine.

ITCS217 Computer Data Communication

Foundation of Data Communication

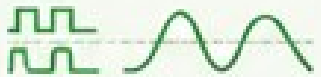
1. Fundamentals (Weeks 1–2)

- Introduction to Data Communications
- OSI and TCP/IP Model
- Network Model and Topology
- Physical Layer



2. Transmission (Weeks 3–6)

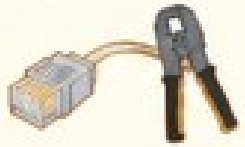
- Data and Signals
- Data Rate Limit
- Transmission Impairment
- Digital Transmission
- Anolog Transmission
- Bandwidth Utilization
- Multiplexxing



3. Media & Multiplexing (Weeks 7–8)

- Transmission Media (Part 1)
- Transmission Media (Part 2)

LAN Cable Crimping Lab (Week 8)



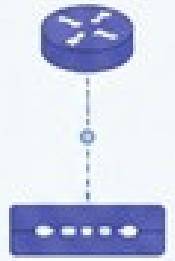
4. Data Link Layer (Weeks 9–13)

- Intro to Data Link Layer
- Flow Control
- Error Control
- Media Access Control (MAC)
- Ethernet
- Telephone Network



5. Networks & Devices (Weeks 14–15)

- Virtual LANs



Learning Outcomes

- Understand basic concepts of data communication and networking
- Explain OSI model and functions of each layer
- Analyze signals, transmission methods, and media
- Apply error control, flow control, and MAC
- Understand Ethernet, VLANs, and network devices

Evaluation

- Quiz #1 (Week 4)
- Quiz #2 (Week 7)
- Crimp LAN Cables Lab (Week 8)
- Midterm Exam (Week 9) 35%
- Quiz #3 (Week 11) (Lec 9–10)
- Quiz #4 (Week 14) (Lec 11–13)
- Final Exam (Week 17) 35%

Exercise 15% (Weeks 1–8, 9–15) | In-Class Quiz 12% (Weeks 4, 7, 11, 14) | LAN Cable Crimping Lab 3% (Week 8) | Midterm 35% (Week 9) | Final 35% (Week 17)

ITCS226 Introduction to Computer Network

Internet Protocols, Routing, and Applications

1. Network Layer (Wk 1–2)

- Overview OSI & TCP/IP
- Intro to Network Layer
- Forwarding & Routing
- SDN & Network Performance
- IPv4 Addressing



2. Routing (Wk 3–5)

- Interior Routing Algorithms
- Exterior Routing Algorithms
- SDN Control Plane and Data Plane



3. Network Applications (Wk 6–9)

- ICMP, DNS, ARP, DHCP
- NAT
- IPv4 & IPv6

Labs: Network Layer & IP Applications



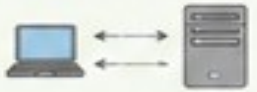
4. Transport Layer (Wk 10–11)

- Transport Layer Overview
- Transport-Layer Protocols (TCP, UDP)



5. Application Layer (Wk 12–14)

- Client-Server & Peer-to-Peer Protocols
- Application Layer Protocols
- Socket Programming, HTTP



6. Advanced Topics (Wk 15)

- Multimedia Networking
- Quality of Service (QoS)



Hands-on Practice

- **Lab 1** Network Command LAB **(Wk 3)**
- **Lab 2** Network Layer Lab Cisco Packet Tracer **(Wk 6)**
- **Lab 3** IP Applications Lab (DNS, DHCP) **(Wk 8)**
- **Lab 4** Cisco Packet Tracer LAB **(Wk 9)**
- **Lab 5** Application Layer Lab (Socket Programming, HTTP) **(Wk 14)**

Evaluation

- **HW#1** (out: Wk 2)
- **Quiz 1** (Wk 3)
- **HW#2** (out: Wk 7)
- **Quiz 2** (Wk 8)
- **Midterm Exam** (Mar 2–9, 2026) **30%**
- **Quiz 3** (Wk 14) (L9–12)
- **HW#3** (out: Wk 12)
- **Final Exam** (May 5–16, 2026) **32%**

In-class Exercise	Homework	Quizzes	Labs	Midterm Exam	Final Exam
10% (1*10)	6% (2*3)	12% (4*3)	10% (2*5)	30%	32%

Learning Outcomes

- Understand Internet architecture and protocols
- Apply IP addressing and routing algorithms
- Explain transport layer protocols (TCP/UDP)
- Use application layer protocols and network applications
- Analyze QoS and multimedia networking

ITCS217 provides the foundation of data communication and data link layer concepts.
ITCS226 builds on that foundation to understand Internet protocols, routing, transport, and applications through theory and practical labs.

Benefits of Computer Data Communication



1 Fast Information Sharing

Data is transmitted instantly across networks.

2 Global Connectivity

Connect with people and systems anywhere in the world.

3 Real-Time Communication

Enable instant messaging, voice, and video communication.

4 Resource Sharing

Share hardware, software, and data efficiently across networks.

5 Collaboration and Teamwork

Work together on projects and share ideas in real time.

6 Remote Work and Online Learning

Access resources, attend classes, and work from anywhere.

7 Support for Business and E-Commerce

Drives business operations, online transactions, and customer engagement.

8 Foundation for Modern Technologies (IoT, Cloud, Cybersecurity)

Enables IoT devices, cloud computing, and protects data and systems.

Computer Data Communications

as Foundation Knowledge for Further Study and Career Paths

Further Study

Cybersecurity
Network security, secure communication, authentication

Internet of Things (IoT)
Smart devices, sensors, edge communication

Cloud & Distributed Systems
Data centers, virtualization, service communication

Wireless & Mobile Networks
Wi-Fi, cellular, mobility

Advanced Networking
Routing, network design, network management

Computer Data Communications Course

- Signals & Media
- Protocols
- Switching
- Routing
- IP Addressing
- Multiplexing
- Performance
- Security Basics

Career Paths

- Network Engineer
- Cybersecurity Analyst
- IoT Engineer
- Cloud / Systems Engineer
- Telecommunications Engineer
- Network Administrator / IT Support

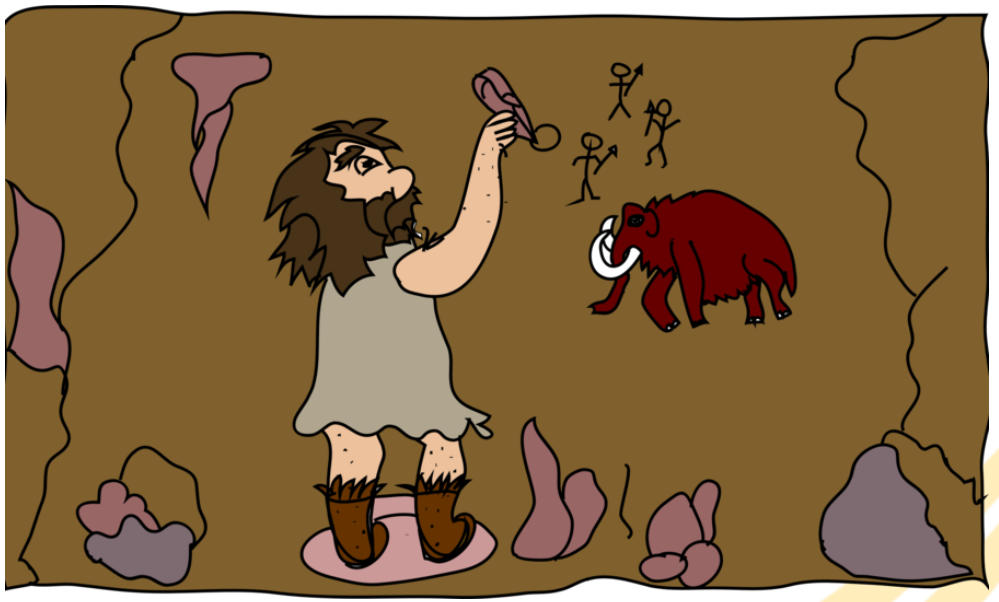
Builds the foundation for advanced study and diverse careers in modern digital technology.

Communication

- People are social beings. They need to communicate in order to gain information and achieve teamwork.

- Early communication is done via speaking, drawing and writing





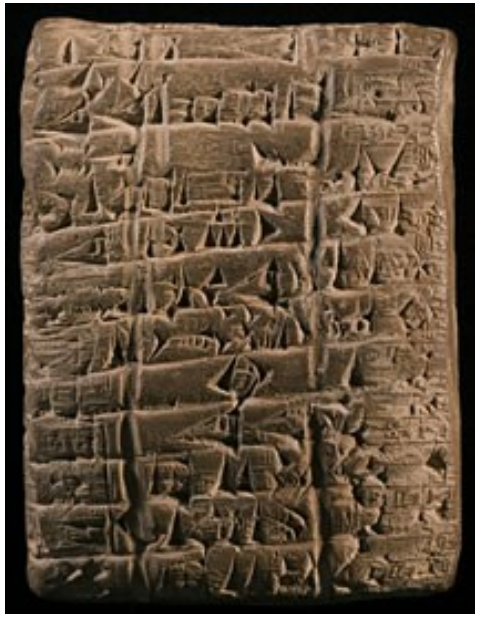
Communication

- **Speaking**
  - Words are sound patterns (symbols).
  - Each word represents a meaning/information (vocabulary).
  - Long connected words need some rules to standardize meaning for complex information. (grammar)
  - The meaningful information can travel back-and-forth through response between the two communication parties form a conversation.
  - Information is stored temporary in human memory.

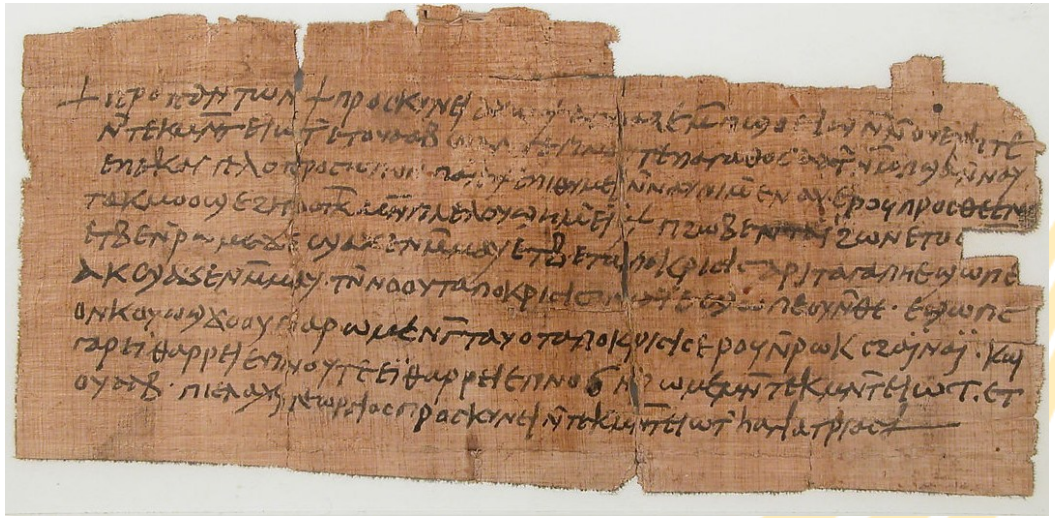
- **Drawing**
  - Drawing represents a meaning/information (sketched image)
  - Information is stored permanently in medium (cave wall, leaves, etc)

Communication

- **Writing**
  - Considered the greatest invention in human history.
    - ✓ Ref: https://tzbarry.com/2018/08/21/invention-writing/
  - Abstracting words into images and simplifying drawing at the same time created writing symbols.
  - Many words can be stored in a medium conveying even more complex meanings which can be stored for a long time.







Communication

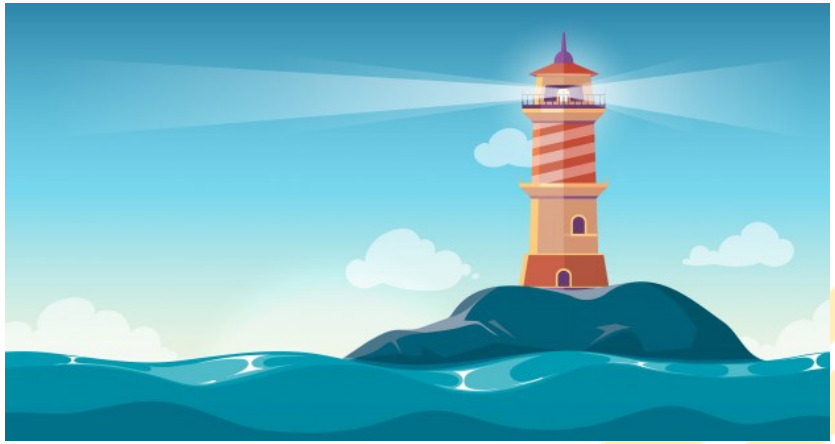
- **Visual signals**







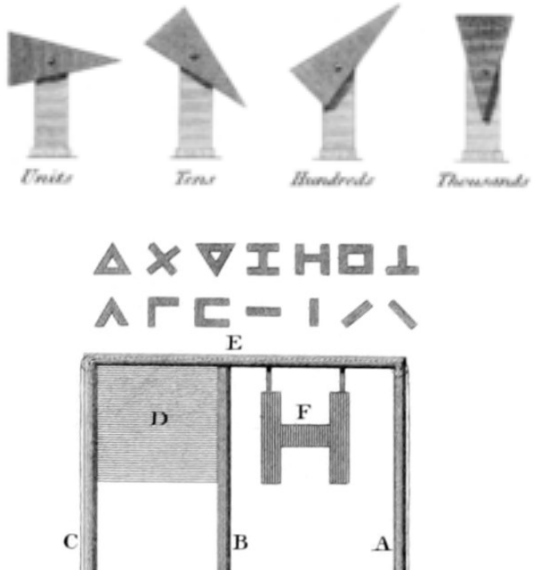




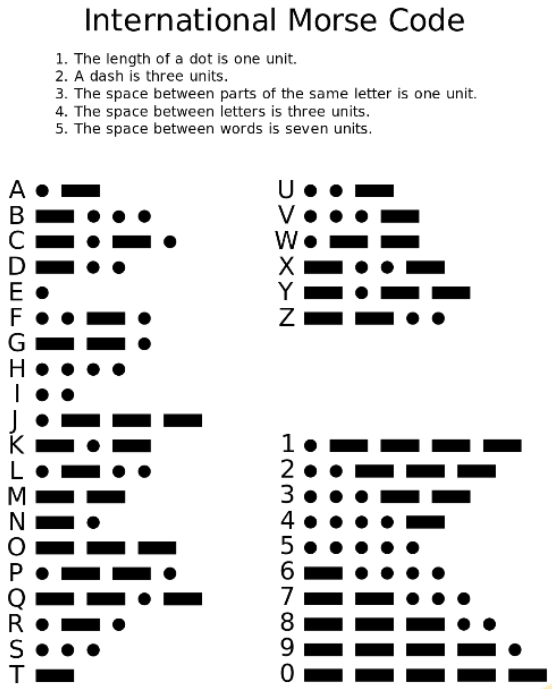
Telecommunication

- **Telecommunication (communication over a long distance)**
  - ✓ Early days, visual and audio signaling can offer long distance communication.
  - ✓ In 1790s, the first semaphore systems emerged in Europe providing visual signaling patterns.
  - ✓ In 1830s, due to the invention of electricity, the electrical telecommunication systems such as telegraph started to appear.





Semaphore systems in Europe

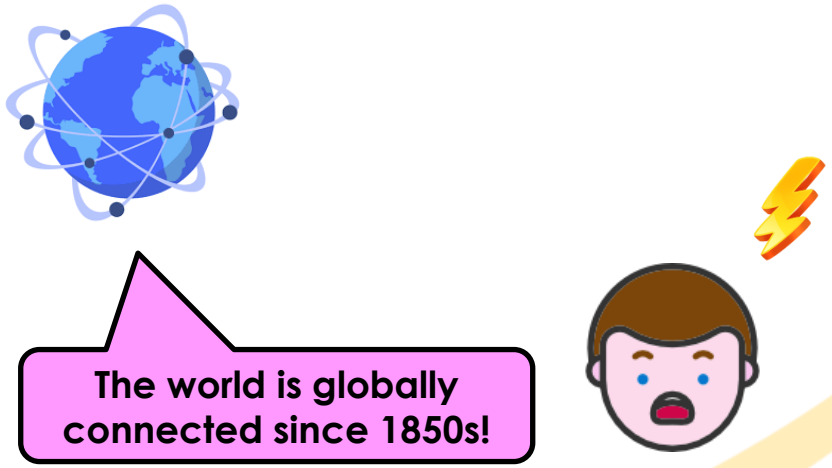


Electrical Telegraph

- Experiments on communication with electricity started in 1726. Scientists including Laplace, Ampère, and Gauss were involved.
  - ✓ An electric current was sequentially applied by the sender through the various wires representing each digit of a message
  - ✓ Different mappings between electrical signal patterns and alphabets were introduced. One of the most successful is **Morse code** (1837) and its application in **telegraph**.
  - ✓ By1851, **telegraph lines** in the United States spanned over 20,000 miles (32,000 kilometers).



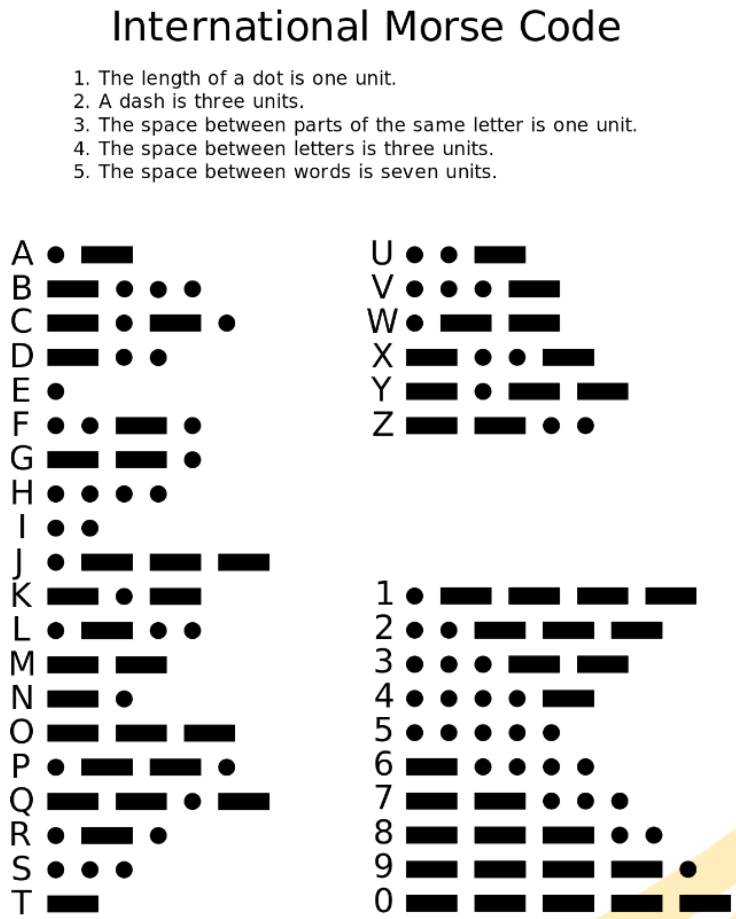
Samuel Morse
(1791-1872)



International Morse Code

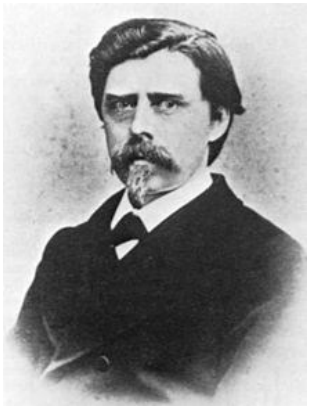
- A good example of light signaling with Morse code is shown here.
  - ✓ https://www.youtube.com/watch?v=aniLPEyp9SA





Multiplexing Telegraph

- Due to the popularity of telegraph, **more people want to send messages at longer distance**. Two major inventions were created after telegraph
  - ✓ **Teleprinter**: (invented by David Edward Hughes in 1855) the transmission and reception of telegraph messages should be automatic without the need for human input. This allows **telegraph** information to be represented as **units of information**.
  - ✓ **Multiplexing**: (invented by Jean-Maurice-Émile Baudot in 1874) the limitation of Morse code was that only one message can be sent at a time, multiplexing allows transmitting more than one telegraph messages simultaneously over a single telegraph wire by using different audio frequencies or channels for each message was introduced.



David Edward Hughes
(1831-1900)



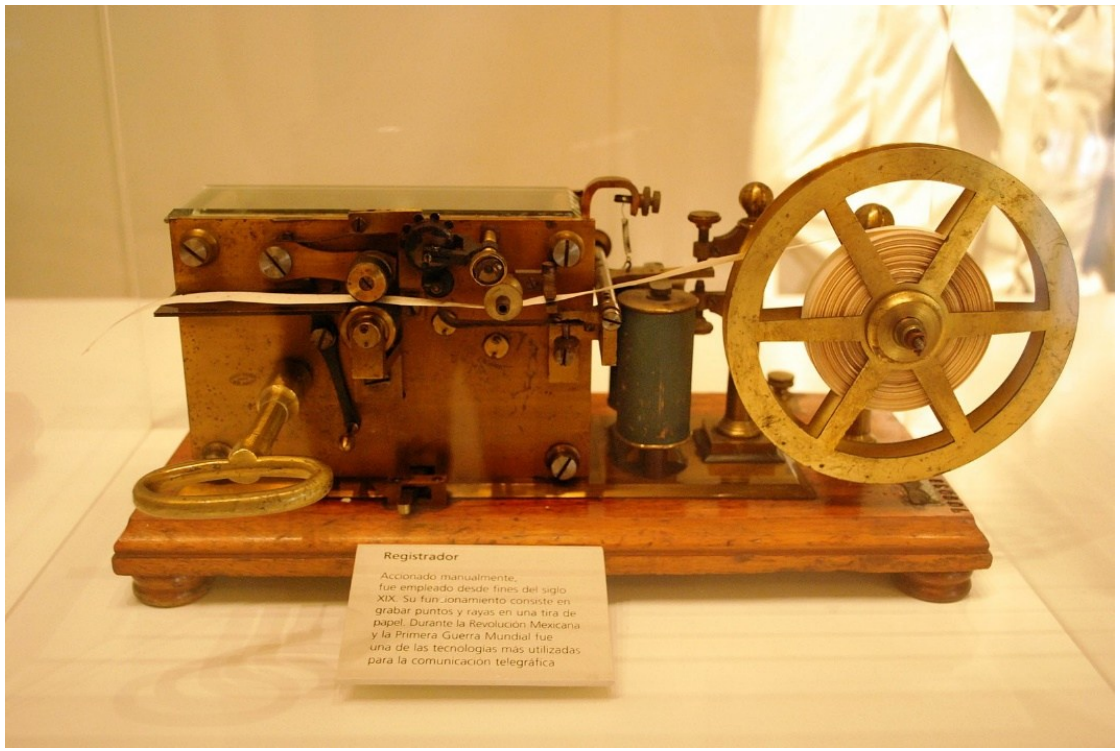
Jean-Maurice-Émile Baudot
(1845-1903)

Hugh and Baudot Telegraph



Hughes's Teleprinter

https://commons.wikimedia.org/wiki/File:Hughes_Teleprinter_-_ACCN_61-47_-_Communication_Gallery_-_BITM_-_Calcutta_2000_203.JPG

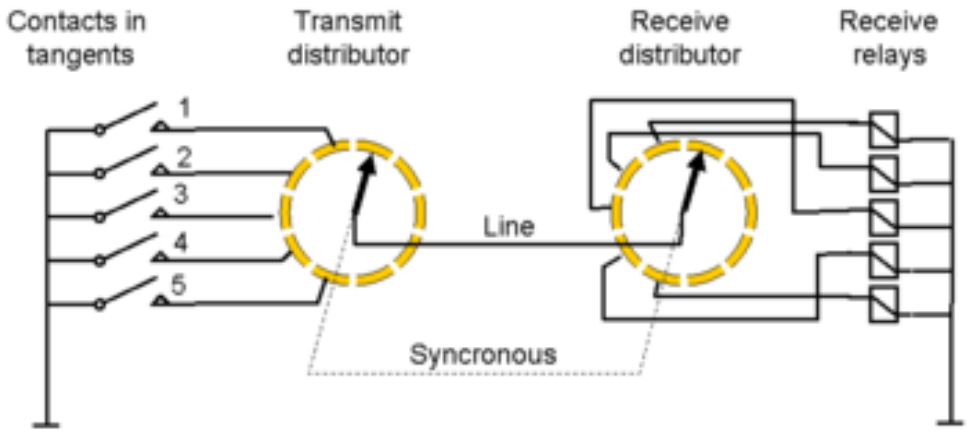


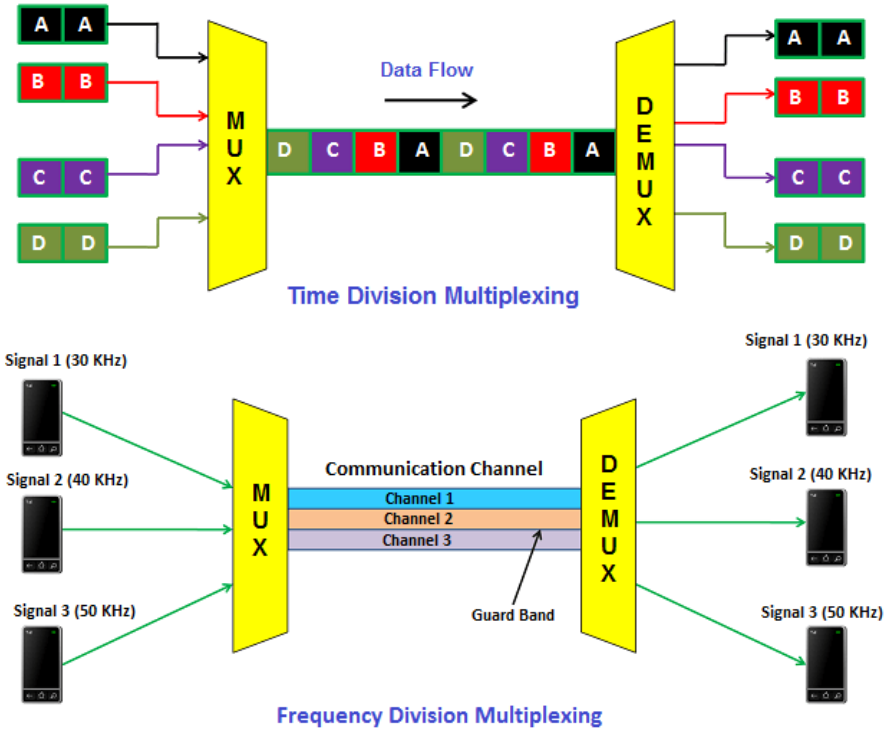
Baudot's Telegraph

https://commons.wikimedia.org/wiki/File:TelegraphPrinter.JPG

Multiplexing

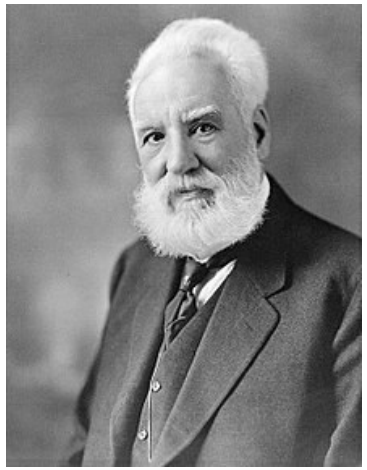
- **Multiplexing is the process of being able to selectively connect an input to a variety of outputs.**
  - ✓ The information is broken into units of information or separate into different line or frequency channels.
  - ✓ Each unit is sent during a time slot (**TDM**: time division multiplexing) or over a single frequency.



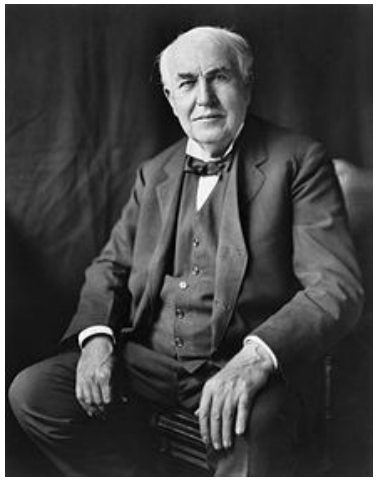


Telephone

- **Baudot's Telegraph** inspires inventors to send more information. The first application (was very much in demand) is to send audio/sound/voice over the line.
- Inventors who worked on the acoustic telegraph included Charles Bourseul, Thomas Alva Edison, Elisha Gray, and Alexander Graham Bell. Their efforts to develop acoustic telegraphy, in order to reduce the cost of telegraph service, led to the invention of the telephone
- The key of **telephone service** is the ability to convert audio/sound waves in the air to electrical signals in **audio frequencies** and transmit the signals via different audio frequency channels (**FDM**: frequency division multiplexing).



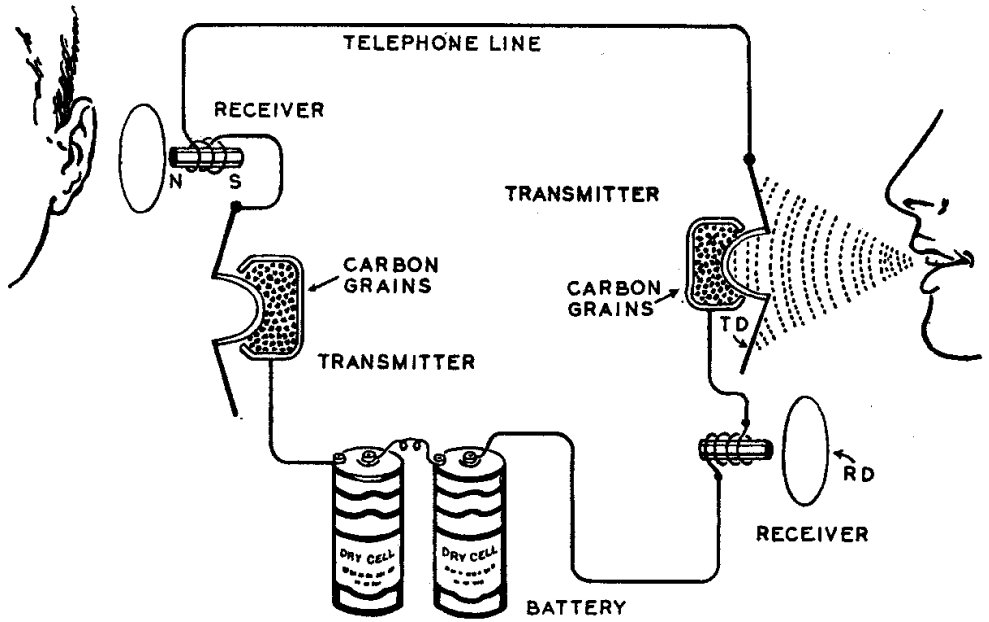
Alexander Graham Bell
(1847-1922)



Thomas Alva Edison
(1847-1931)

Telephone

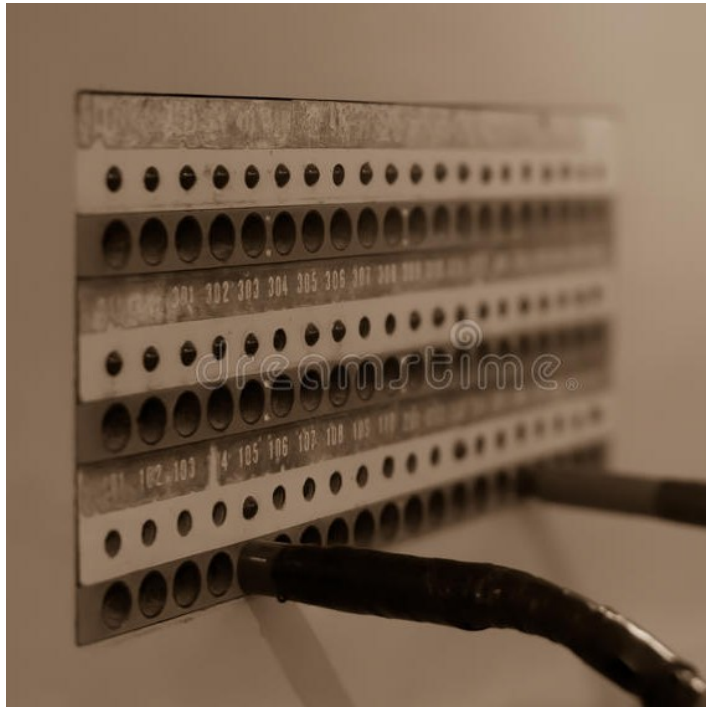
- Early design of telephone systems consists o
  - ✓ Sound wave to electrical signal converter (in audio channels)
  - ✓ Electrical transmitter (via telephone line)
  - ✓ Multiplexer (central management system)
  - ✓ Electrical receiver
  - ✓ Electrical signal to sound wave converter
- Audio signals will be converted to electrical signals which are sent over the lines.

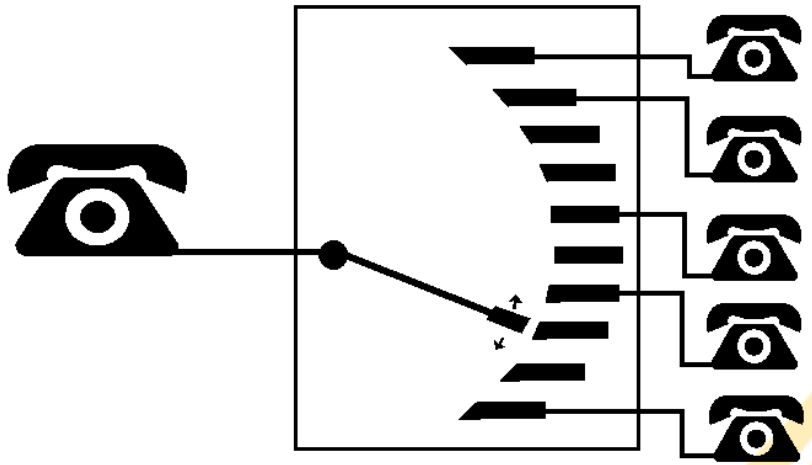


Telephone Switching

- In the early days, analog audio signals cannot be converted to digital bits of information yet. Connecting two telephone parties together required a central office which manually attached two wires together to make a connection.



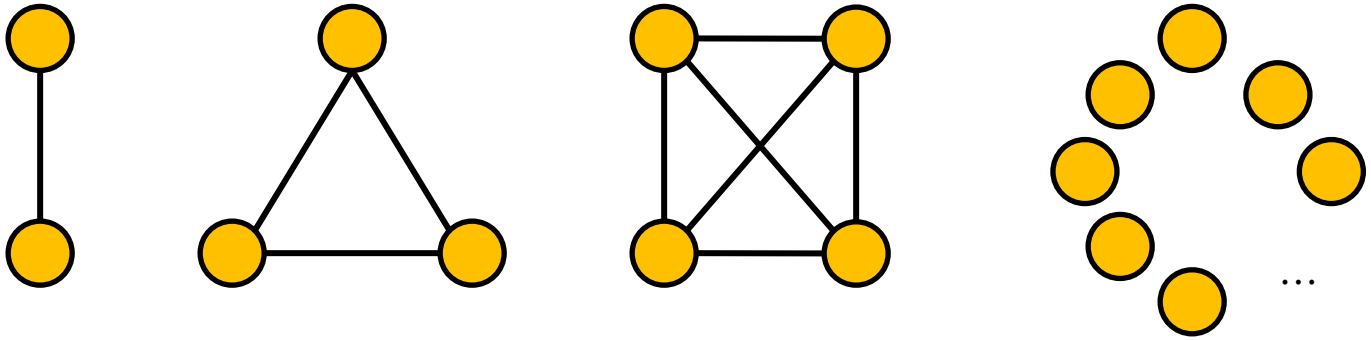




https://www.dreamstime.com/photos-images/telephone-switch-old.html

Telephone Switching

- Assuming a full mesh (connecting all **n** telephones directly together)
- Each customer can call any other (**n-1**) customer.
- How many lines are required to connect each one customer to other **n-1** customers?



n = 2
links = 1

n = 3
links = 3

n = 4
links = 6

n = n
links = ?

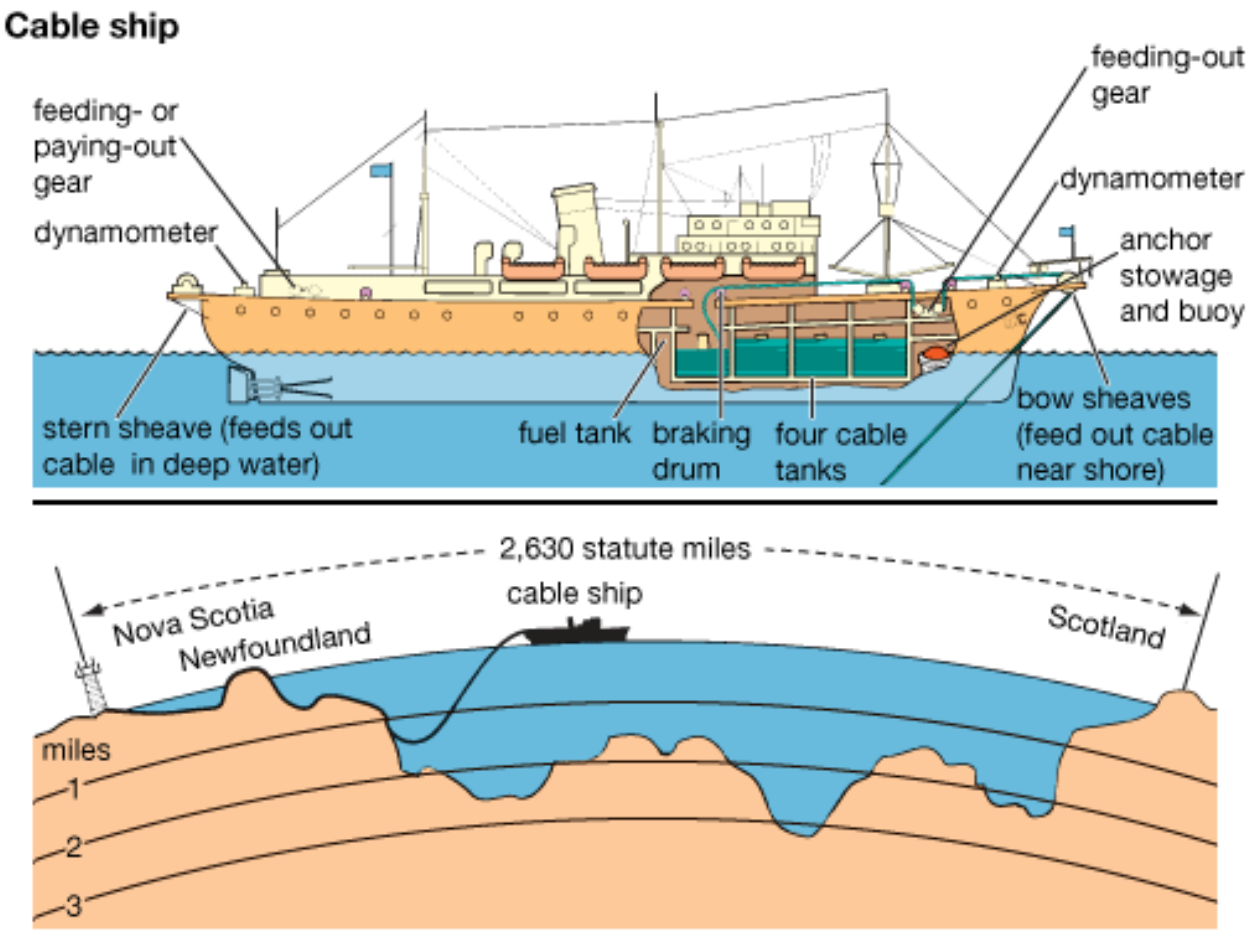
$$\#\ links = \sum_{i=1}^{n-1} i = \frac{n(n-1)}{2}$$

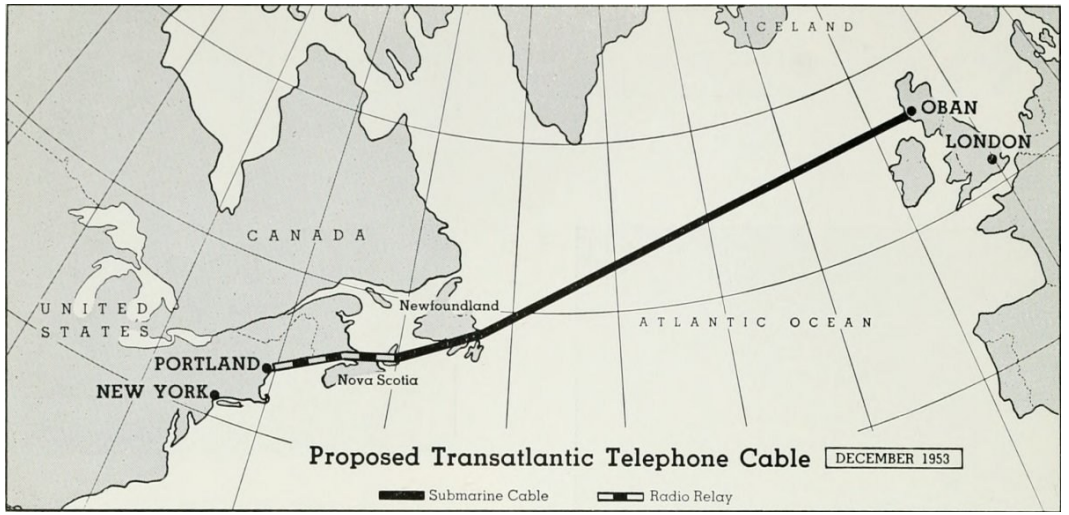
- **Questions**: scalability?
- **Solution**: Multiplexing is needed.



Long Distance Telephone

- The first transcontinental telephone call occurred on January 25, 1915.







Wireless Telecommunication

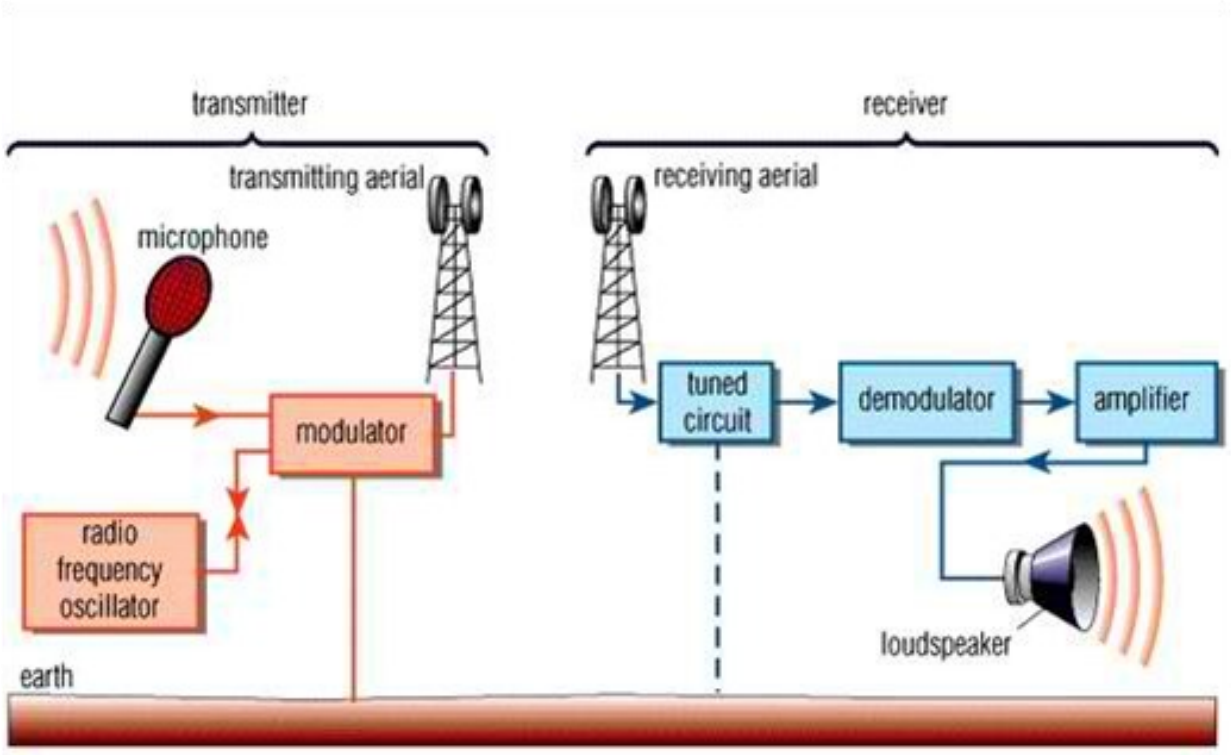
- James Clerk **Maxwell**, in the 1860s, predicted the existence of electromagnetic waves, which laid the theoretical foundation for wireless communication.
- Heinrich **Hertz**, in 1888, experimentally confirmed Maxwell's predictions by demonstrating the existence of electromagnetic waves. He was the first to transmit and receive radio waves in a controlled environment, proving that electromagnetic waves could propagate through space.
- Guglielmo **Marconi** developed Practical Radio Transmission and Wireless Telegraphy in 1897.
- **Alexander Graham Bell** developed wireless telephone in 1946.
- GSM and CDMA, base technology of Cellular telephone, were introduced in 1992 and 1993.
- 3rd Generation mobile phone (Internet through broadband wireless) was introduced in 1997.
- Wireless LANs using 2.4 GHz spectrum was introduced in 1999.



Heinrich Hertz
(1857-1894)

Wireless Telecommunication

- In order to transmit information wirelessly, the information has to be converted to electromagnetic waves through modulators.
- The electromagnetic waves can be converted back to become electrical information using a demodulator.



Wireless Revolution

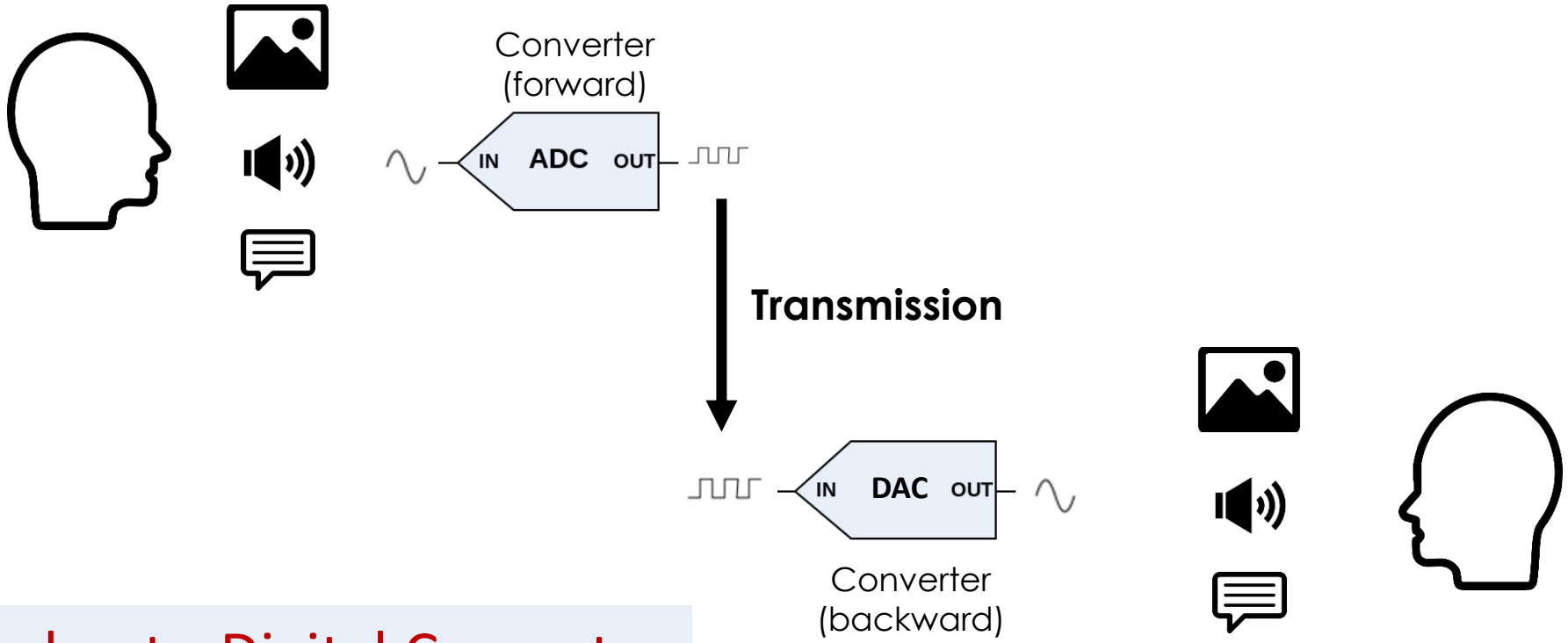
- The wireless revolution began in the 1990s with the advent of digital wireless networks leading to a **social revolution**, and a paradigm shift from wired to wireless technology including the proliferation of commercial wireless technologies such as **cellular phones, wireless computer networks, satellite networks.**

- The wireless revolution has been driven by advances in radio frequency (RF) and microwave engineering, and the transition from analog to digital RF technology, which enabled a substantial increase in voice traffic along with the **delivery of digital data such as text messaging, images and streaming media**

- Wireless computer networks and cellular networks together with access to Internet greatly increase popularity and ability of the Internet and this leads to the current boom of the Internet.

Telecommunication Concept

- The concept of telegraph, telephone, radio and television are still being used in modern time with communication through cellular and satellite communication or even the Internet.



Analog-to-Digital Converter

The Internet

- The widespread use of telephone and digital computer in 1950s allows inventors to combine computer technology with telephony which enable computers to send/receive data to/from other computers.
- Paul Baran proposed a distributed network based on data in message blocks in the early 1960s and Donald Davies conceived of **packet switching** in 1965 at the National Physical Laboratory (NPL) in the UK and proposed building a national commercial data network.
- Today packet switching is the dominant basis for data communications in computer networks worldwide. **It was key to the development of the Internet**.



Donald Davies
(1924-2000)

Circuit and Packet Switching

- The switching used **in telephone network** is called **circuit switching** where the two communication channels or links have to be *dedicatedly connected*.
  - ✓ All information between the sender and the receiver has to go through a single connected link (a circuit).

- **Packet switching** allows information to be broken up into bits (similar to multiplexing telegraphy) and the delivery of bits (data streams) realized as sequences of packets over a computer network using statistical multiplexing or dynamic bandwidth allocation techniques.

- The theory of packet switching is based on the concept of **queueing theory** developed by **Leonard Kleinrock**.
  This is **the foundation of the Internet** we use today.



Leonard Kleinrock
(1934-present)

ARPANET

- ARPANET (Advanced Research Projects Agency Network) is the precursor to the Internet. The ARPANET was established by the **Advanced Research Projects Agency (ARPA)** of the United States Department of Defense in order to access remote computers and control machines around the world.

- ARPANET is the first wide-area **packet-switching network** with distributed control and one of the first networks to implement the TCP/IP protocol suite.

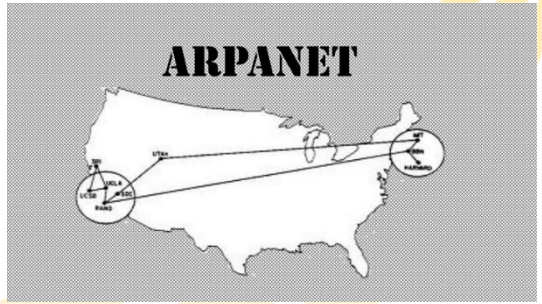
- The first computers were connected in 1969 and the Network Control Program was implemented in 1970.

The Rise of the Internet

- Further software programs were developed on top of ARPANET including remote login, file transfer and email. The network expanded rapidly and was declared operational in 1975 when the control of ARPANET was passed to the Defense Communications Agency.

- Version 4 of **TCP/IP suite** was installed in the ARPANET for production use in January 1983 after the Department of Defense made it **standard** for all military computer networking.

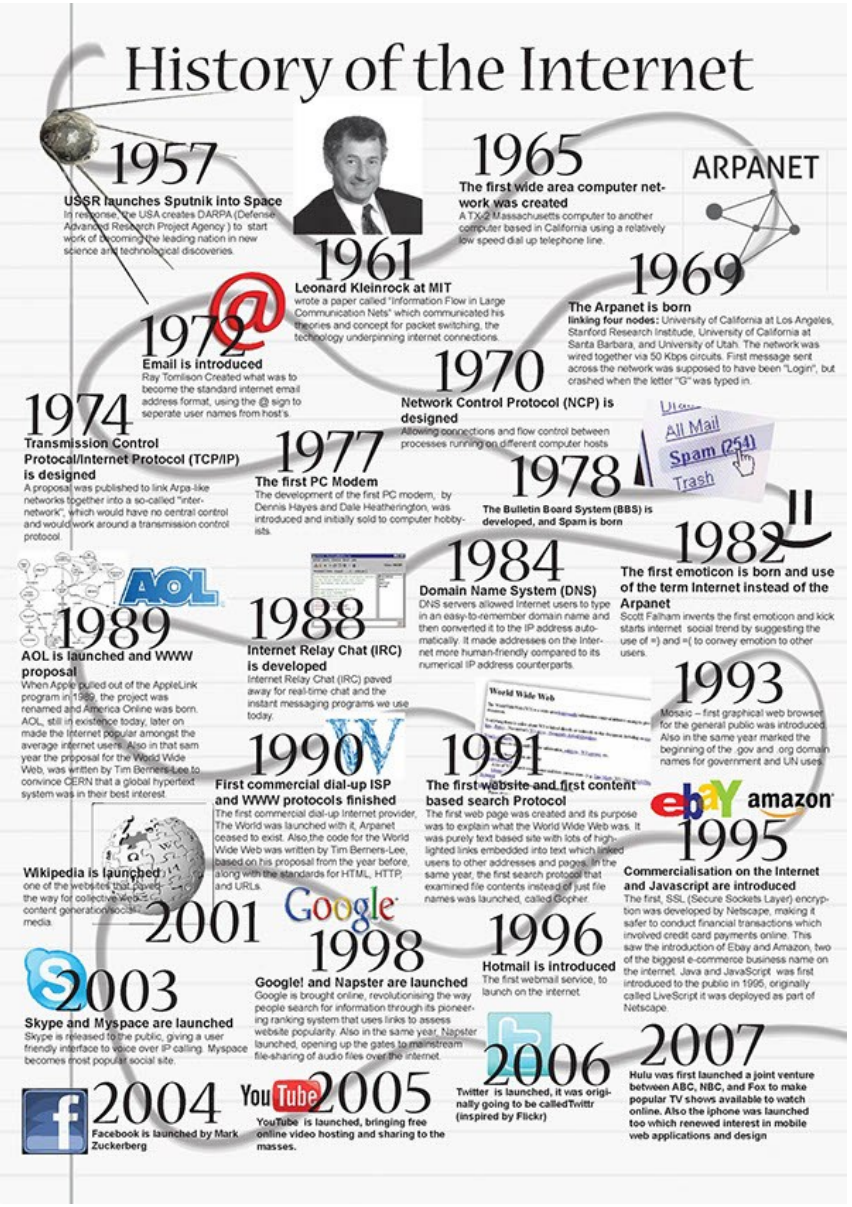
- The ARPANET project was formally decommissioned in 1990, after DARPA partnerships with the telecommunication and computer industry had paved the way for the widespread adoption of the Internet protocol suite as part of the private sector expansion and commercialization of a world-wide network, known as **the Internet** (the interconnection of electronic computers using TCP/IP as the communication protocol).

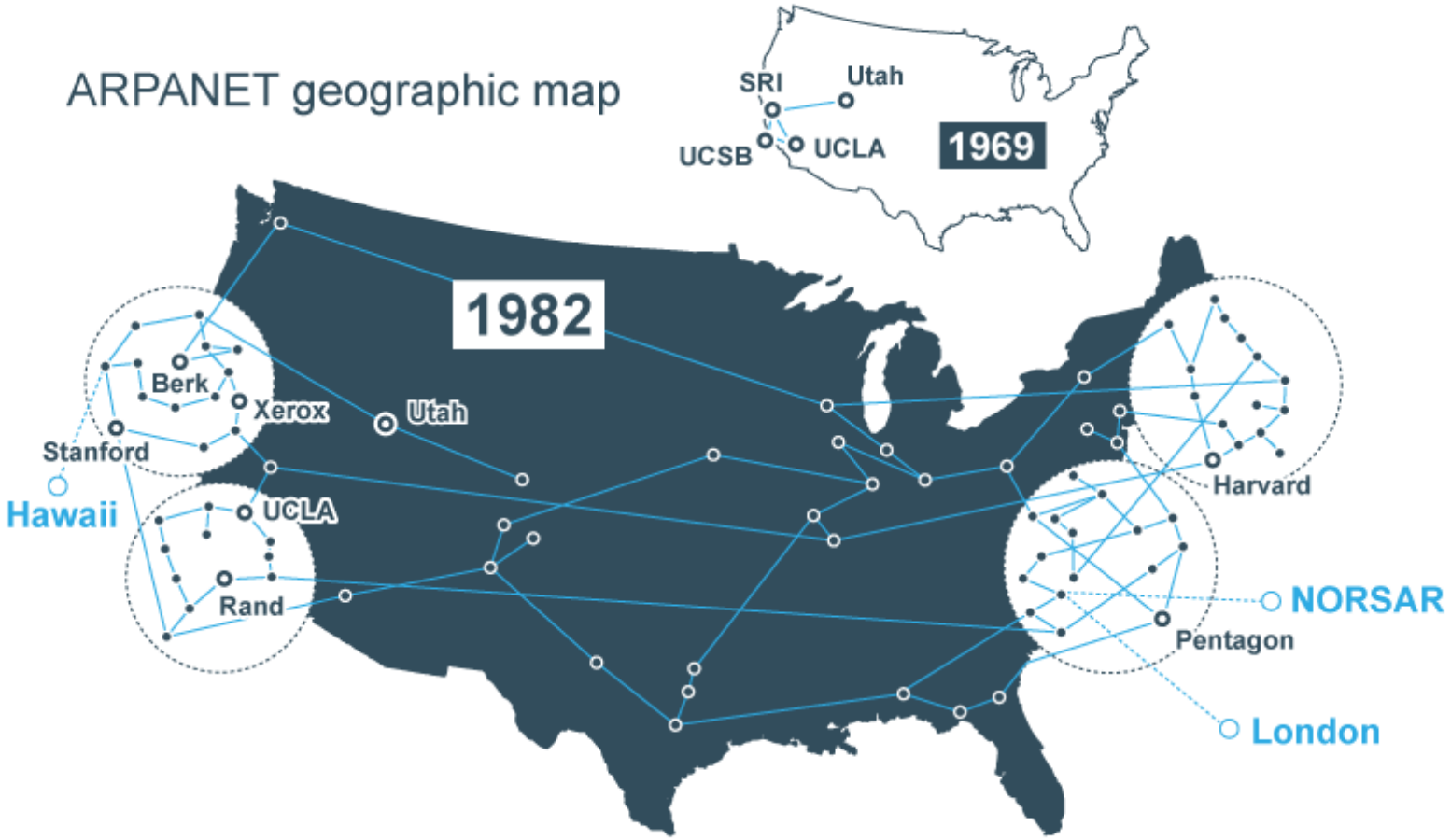


The Internet

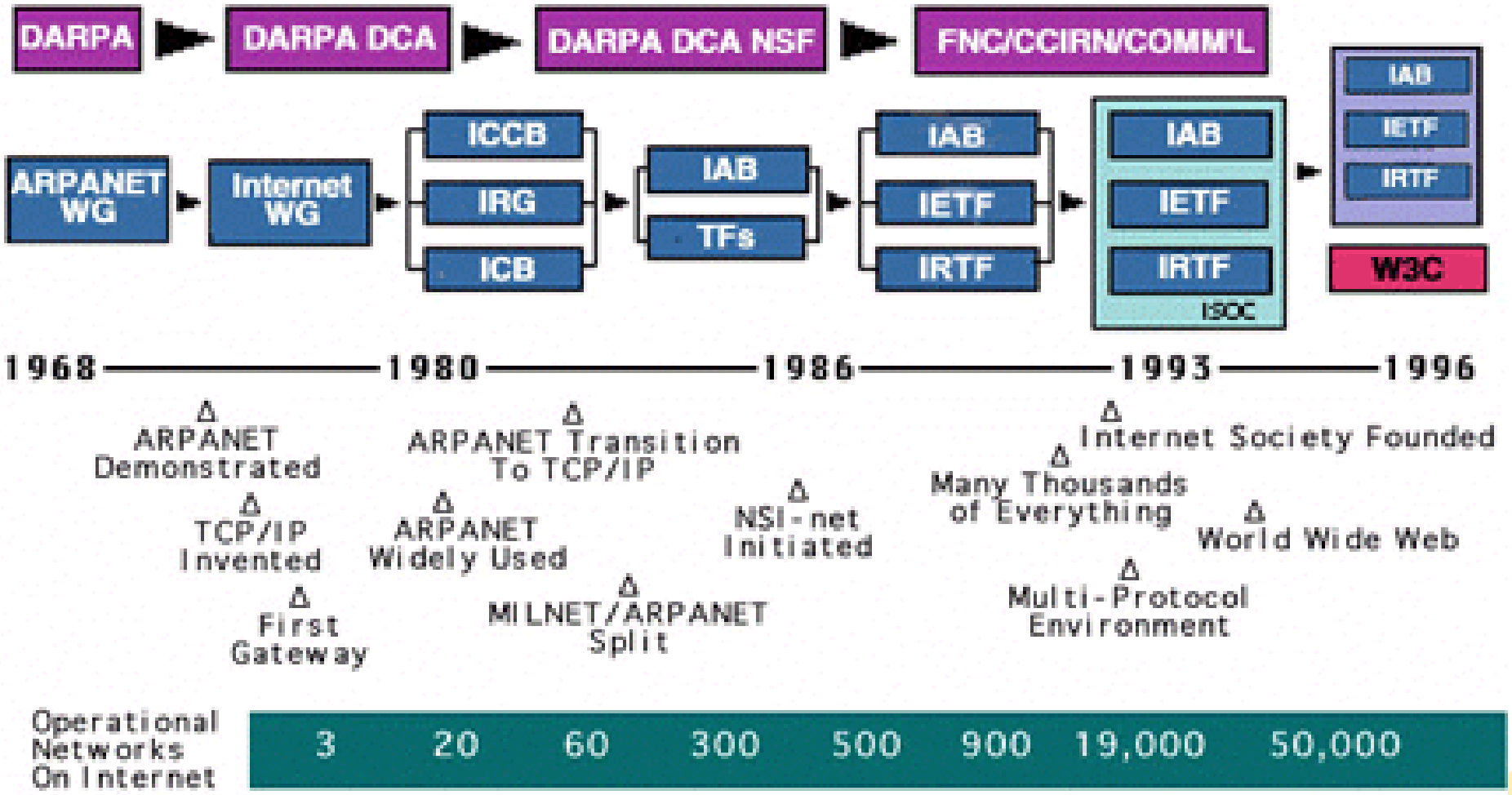
- The Internet was first an interconnection of routers called ARPANET used for military purpose in 1960s.
- Since then, more routers are added from around the world and computers are connected to these routers.
- Now, there is no single owner of the Internet, but rather multiple owners: multiple Internet Service Provider (ISP) corporations like AT&T and Verizon, etc.

The History of the Internet



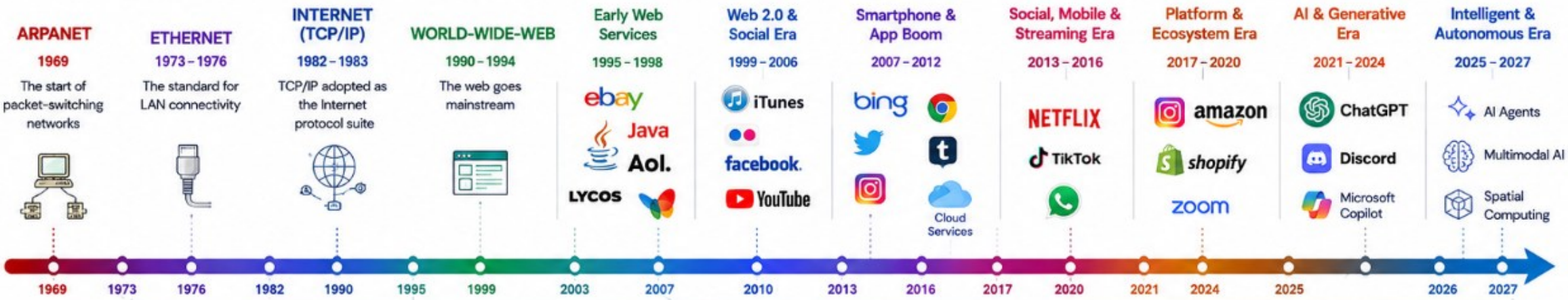


The Organizations of the Internet



The Evolution of Internet Applications (1969 – 2027)

From Research Networks to Intelligent, Connected, and Generative Experiences



INCUMBENTS

Knowledge	Communication	Commerce	Entertainment	Productivity
Google	facebook.	amazon	YouTube	Gmail
		Booking.com	NETFLIX	slack
		Uber	TikTok	zoom
		DOORDASH	ROBLOX	shopify
		airbnb		

NEW PLAYERS

Personal Assistant (LLM Chatbots)	Personal Assistant (Conversational)	Personal Assistant (Travel)	Personal Assistant (Work & Productivity)	Personal Assistant (Creative)
ChatGPT	character.ai	WANDERBOT	character.ai	Plus
Bard	Inflection	Shopping assistant	Eleven Labs	yoodli
Jasper	Replika	Returns manager	synthesia	Co-pilot for content
		Travel planner	AI companion	Email manager
				Calendar manager

- **Connecting People** — Global communication and community building
- **Sharing Knowledge** — Open access to information and education
- **Enabling Commerce** — Digital markets and platform economies
- **Empowering Entertainment** — On-demand content and interactive experiences
- **Driving Productivity** — Tools that improve how we work and collaborate
- **Shaping the Future** — AI, automation, and new digital frontiers

What Comes Next (2025–2027)

- **AI Agents & Automation** — Autonomous agents helping with complex tasks
- **Multimodal AI** — Text, voice, image, video understanding & creation
- **Spatial Computing** — AR/VR/MR experiences blending digital & physical
- **Hyper-Personalization** — Smarter, context-aware services for everyone

The Internet continues to evolve — from connecting computers to connecting people, knowledge, and intelligence.
The next frontier: AI-powered, human-centered, and sustainable digital experiences.

Research Era · Foundation Era · Web Era · Social Era · Mobile Era · Platform Era · AI Era · Future Era

Exercise 1

- Which of the following is **NOT** an appropriate meaning of data communication?

A. Exchanging data between two or more connected devices.
B. Processing data entirely within one isolated computer system.
C. Transmitting information through a communication medium
D. Delivering data accurately to an intended destination.
E. Sharing digital information according to agreed communication protocols.

Exercise 2

- Which statement best describes wireless communication?

A. Data is transmitted only through copper cables.
B. Data is carried by electromagnetic waves without a physical wired medium.
C. Data can be transmitted only by visible light.
D. Data communication does not require encoding or modulation.
E. Data is transmitted without using any electromagnetic wave or signal.

Exercise 3

- Which technology is mainly designed for short-range personal-area communication rather than long-distance telecommunication?

A. Radio broadcasting
B. Television broadcasting
C. Telegraph
D. 5G cellular network
E. Satellite communication
F. Telephone network
G. Bluetooth

Exercise 4

- Assume a full-mesh connection is used in an old telephone switching system. If there are 10 customers, how many direct lines are required to connect every customer to every other customer?

- Hint: Number of links = $n(n - 1) / 2$

A. 100
B. 90
C. 45
D. 11
E. 10

Exercise 5

- Which type of switching is used as the main basis of the Internet?

A. Circuit switching
B. Packet switching
C. Signal switching
D. Package switching
E. Multiplex switching

Exercise 6

- Which statement is **NOT** correct?

A. ARPANET was developed by the Advanced Research Projects Agency of the United States Department of Defense.

B. ARPANET was the first wide-area circuit-switching network developed for communication between computers.

C. ARPANET was an important technological precursor to the development of the modern Internet.

D. ARPANET was one of the first networks to implement the TCP/IP protocol suite.

E. ARPANET initially connected research institutions and universities across different geographical locations.

Exercise 7

- Which type of switching was mainly used by the traditional telephone network?

A. Circuit switching
B. Packet switching
C. Signal switching
D. Package switching
E. Internet switching

Exercise 8

- Which statement best distinguishes telecommunication from data communication?
  - A. Telecommunication covers information exchange over distance, while data communication focuses on digital data between devices.
  - B. Telecommunication uses only analog signals, while data communication always uses digital signals between computers.
  - C. Telecommunication requires telephone networks, while data communication requires only local computer network connections.
  - D. Telecommunication supports voice communication, while data communication cannot transmit audio or video information.
  - E. Telecommunication operates without protocols, while data communication always depends on network communication protocols.

Exercise 9

- Which event helped Internet applications become widely accessible to general users?
  A. The development of Ethernet for local area network communication.
  B. The introduction of the World Wide Web and web browsers.
  C. The creation of packet switching for research communication networks.
  D. The adoption of telephone switching for long-distance voice services.
  E. The invention of copper cables for digital signal transmission.

Exercise 10

- Which development best represents the mobile and cloud era?

  A. Applications became limited to desktop computers inside research laboratories.

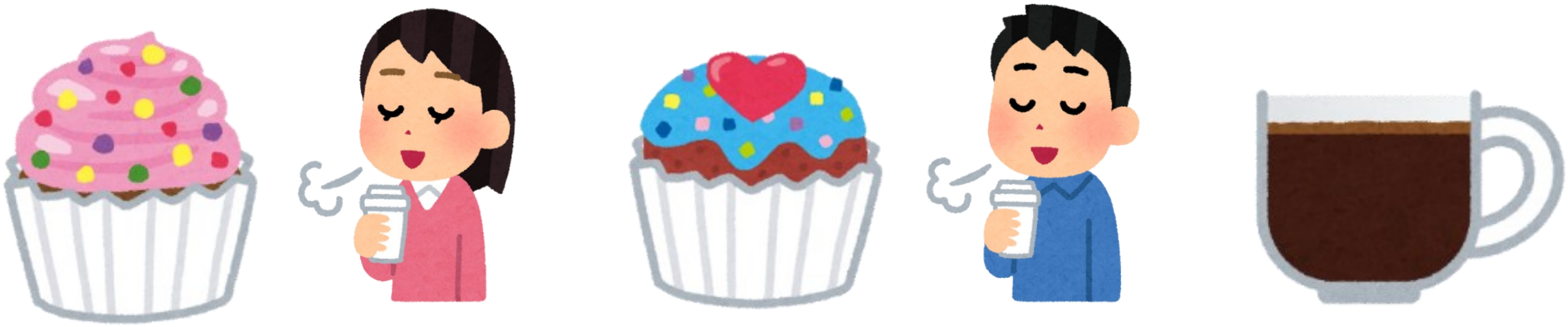
  B. Applications became accessible across many devices, with data and services available from anywhere.

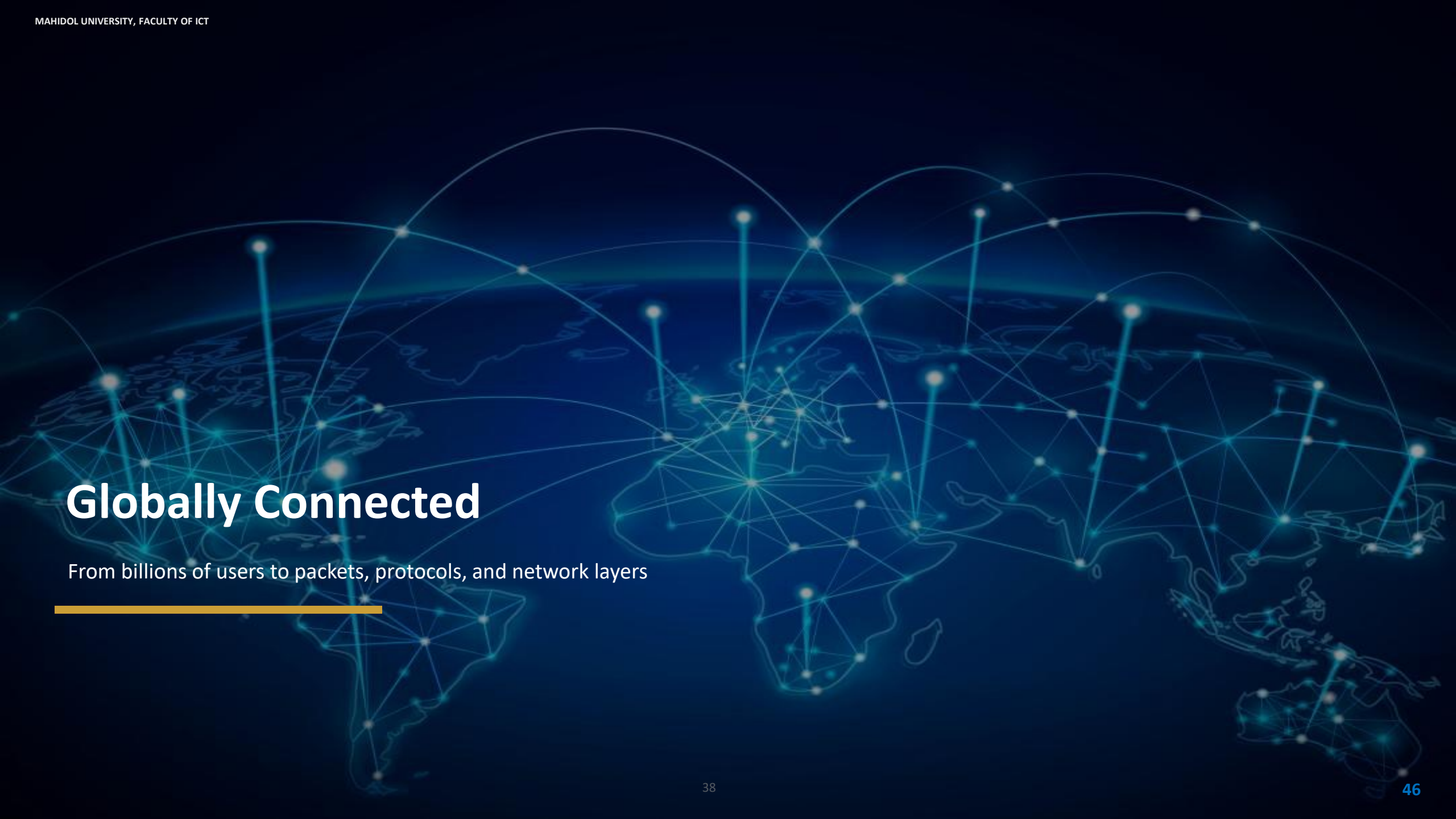
  C. Applications stopped using Internet protocols for communication between devices.

  D. Applications returned to isolated systems without remote data access.

  E. Applications depended only on local storage and offline software installation.

Let’s have a Break!





Globally Connected

From billions of users to packets, protocols, and network layers

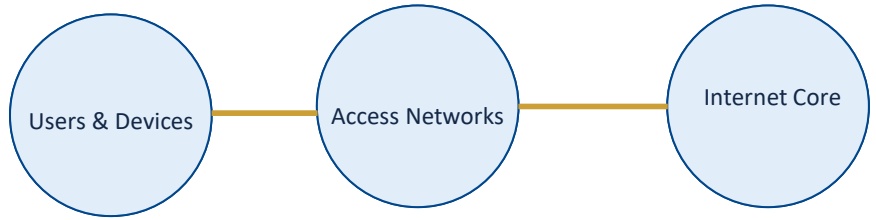
MAHIDOL UNIVERSITY, FACULTY OF ICT

Global Digital Overview 2026

Digital services depend on global-scale data communication infrastructure

8.25B	6.04B	5.78B	5.66B
people on Earth	internet users worldwide	unique mobile users	social media user identities

- More users and devices create more packets, sessions, and traffic flows.
- Internet scale requires addressing, routing, multiplexing, and layered **protocols.**
- Performance and reliability now affect education, business, healthcare, and public services.



https://datareportal.com/reports/digital-2026-global-overview-report
https://www.reuters.com/business/media-telecom/indias-vast-internet-social-media-apps-market-2026-01-29/

Source: DataReportal Digital 2026 Global Overview Report (Oct 2025 data).

MAHIDOL UNIVERSITY, FACULTY OF ICT

Thailand: A Highly Connected Digital Society

Connectivity indicators show why data communication is a foundation for modern services

71.6M	67.8M	96.6M	56.6M
population	internet users (94.7% penetration)	cellular mobile connections (~135% of population)	social media user identities (79.1%)

- Thailand is mobile-first: many users access services through cellular and Wi-Fi networks.
- High internet penetration increases expectations for always-on, low-delay communication.
- Digital services require dependable wired, wireless, cloud, and edge connectivity.



Course link: from connected users to packets, frames, signals, and protocols

Source: DataReportal Digital 2026: Thailand (Oct/End-2025 data).

MAHIDOL UNIVERSITY, FACULTY OF ICT

What Data Communication Must Solve

The rest of the course explains how networks move data correctly, efficiently, and securely

Capacity

How much data can the channel carry? Bandwidth, multiplexing, throughput

Delay

How long does delivery take? Latency, queueing, congestion

Reachability

How does a packet find a destination? Addressing, routing, switching

Trust

How can communication be reliable and safe? Error control, authentication, encryption

Learning path:

Transmission media → Signals → Framing → Switching & Routing → Protocols → Internet applications

ITCS217 Computer Data Communication **ITCS226 Computer Network**

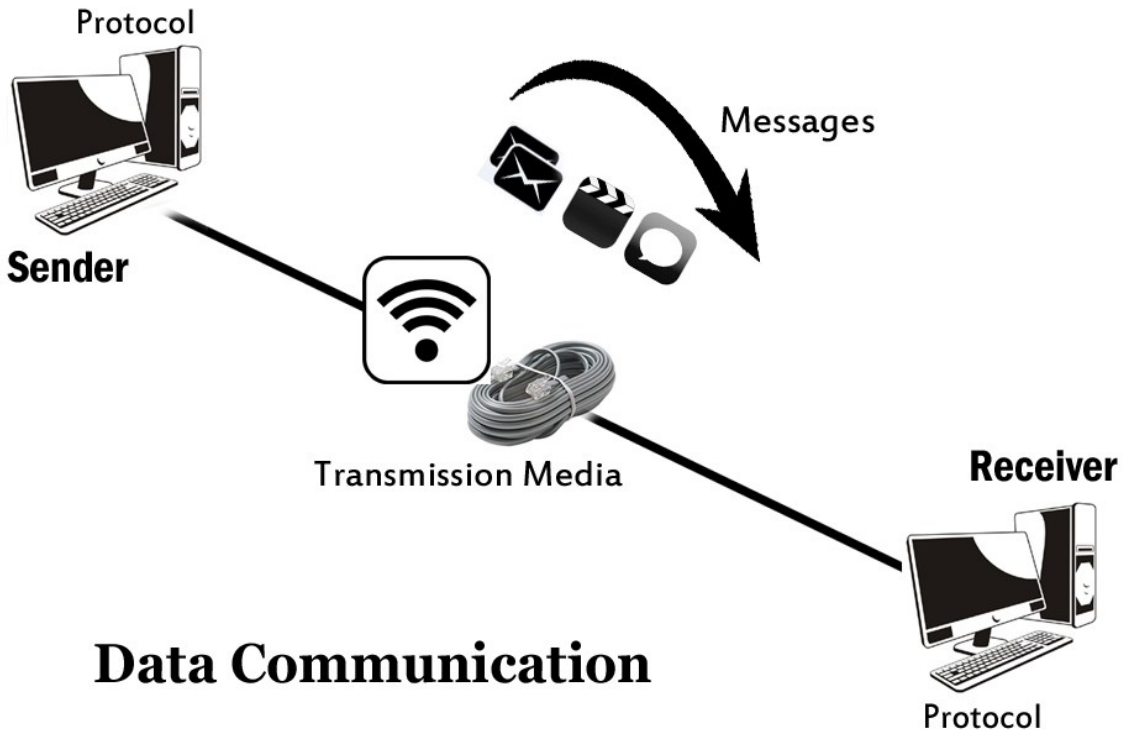
Updated section summary: replaces previous Slides 38–46 with four focused slides.

How does the Internet Work?

- **The Internet** carries data from one computer to other computers in the network using a specific standardized method (**TCP/IP protocol suite**).
- The Internet works through a packet routing network in accordance with the Internet Protocol (IP), the Transport Control Protocol (TCP), User Datagram Protocol (UDP) and other protocols in the suite.
- **Protocol**
  - A set of rules specifying how computers should communicate with each other over a network.
- **Packet**
  - A data container flows inside the Internet.
- **Routing**
  - A way to direct and pass packets from source to destination along a path.

Data Communication Model

- Data communication consists of 5 elements
  - Sender
  - Receiver
  - Message
  - Protocol
  - Transmission media



Data Communication

Data Communication

- **Definition**:
  - **Data communications** are the exchange of data between two devices via some form of transmission media.
  - **Data**
    - (for Computer people) data are the quantities, characters, or symbols on which operations are performed by a computer, being stored and transmitted in the form of electrical signals and recorded on magnetic, optical, or mechanical recording media.
    - (For normal people) data are facts and statistics collected together for reference or analysis.

Communication Protocol

- **A communication protocol** is a system of rules that allow two or more entities of a communications system to transmit information via any kind of variation of a physical quantity.
- A protocol is analogous to "communication language" in human terms
- The protocol consists of
  - **Syntax**: rules of communication (e.g. grammar)
  - **Semantics**: means of symbols (e.g. vocabulary)
  - **Synchronization**: timing of conversation (e.g. timing) and error recovery methods

Syntax	Grammar
Semantics	Vocabulary
Synchronization	Conversation



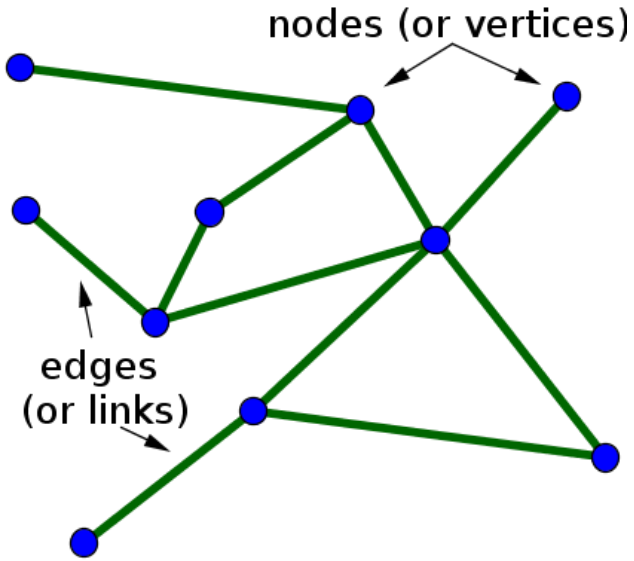
Computer Communication Protocol



Human Communication Language

Network

- **Definitions**:
  - Network is
    - an arrangement of intersecting horizontal and vertical lines.
    - a group or system of interconnected people or things.
- A network consists of **nodes** and **edges**.
- A network is not the Internet. The Internet is a type of network that uses TCP/IP protocol.

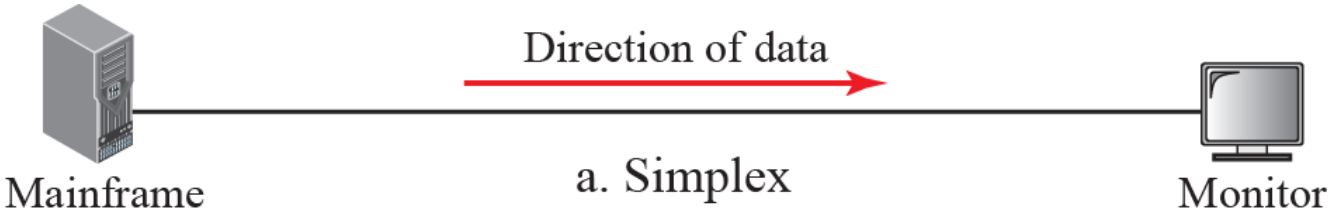


Node: a communication device/human

Edge: a direct communication path between two nodes

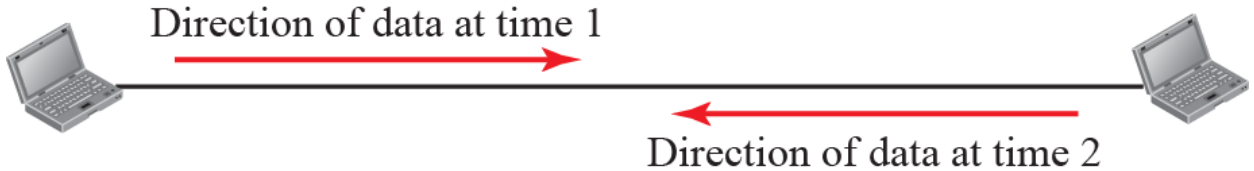
Type of Communication

- Based on flow of data



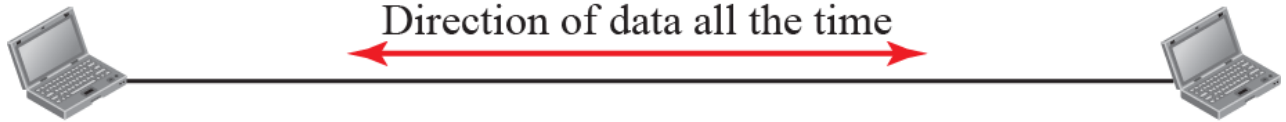
a. Simplex

Television, AM/FM Radio, Newspaper, LINE notify



b. Half-duplex

human conversation, walkie-talkie, Delivery truck

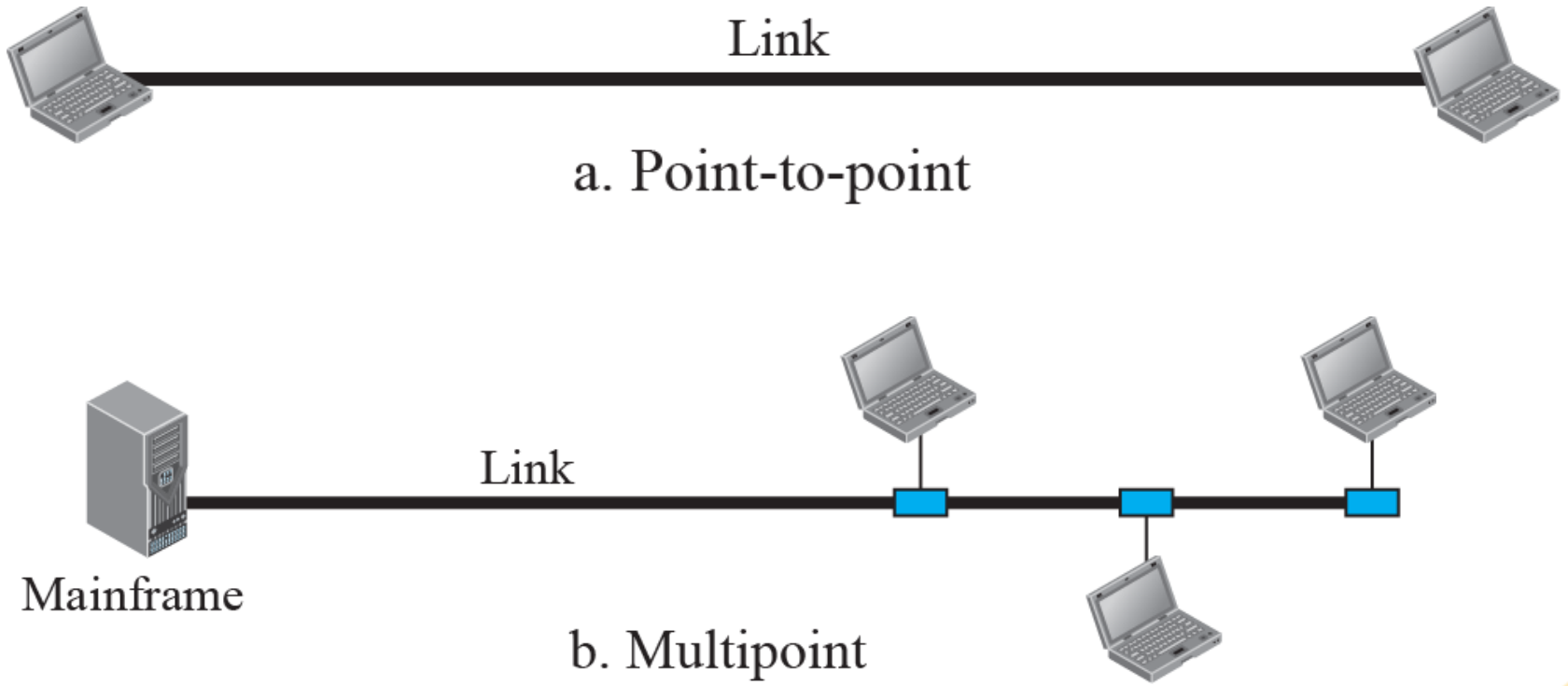


c. Full-duplex

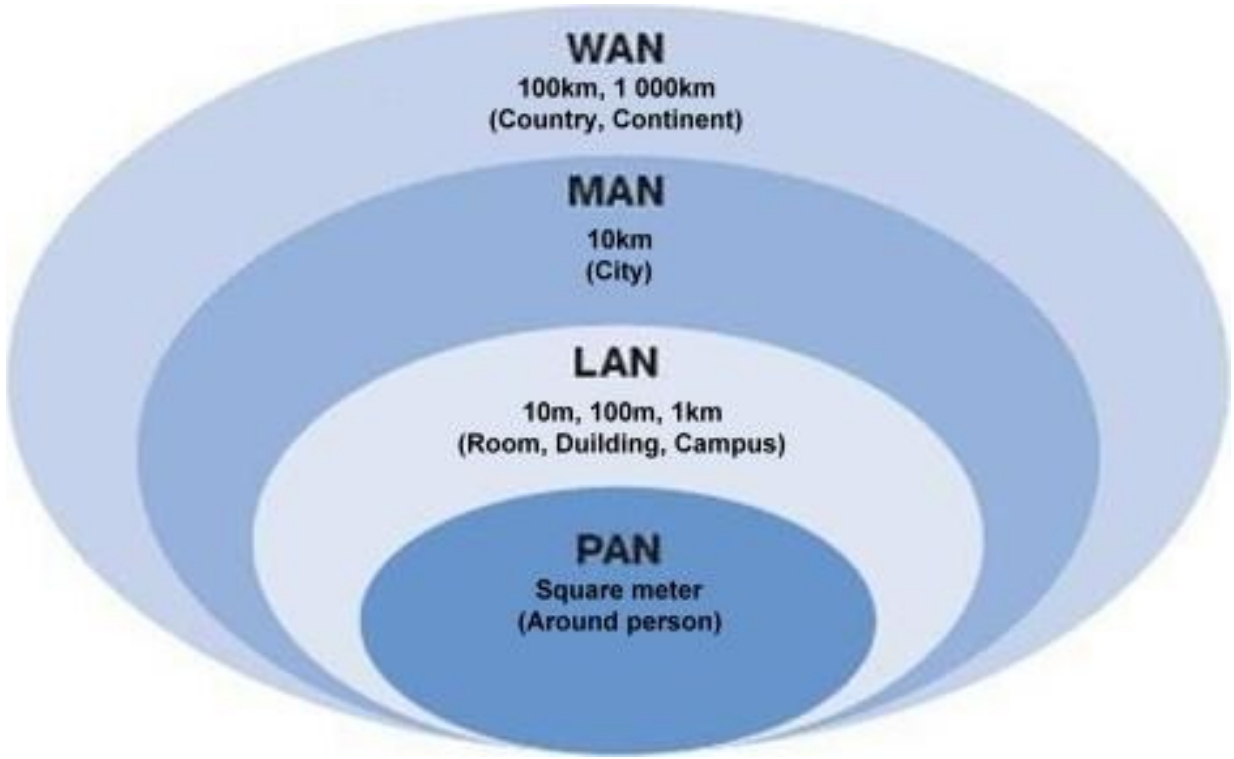
Modern computer communication, 4G/5G cellular, Wi-Fi

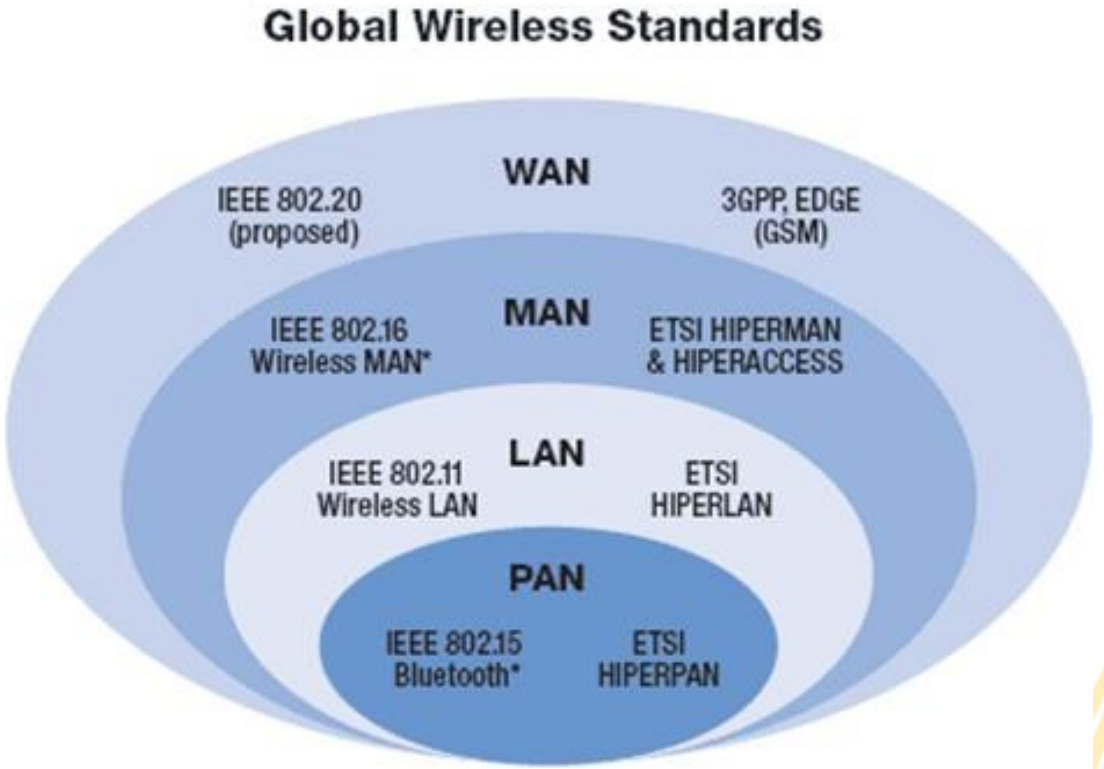
Types of Connection

- Based on connectivity



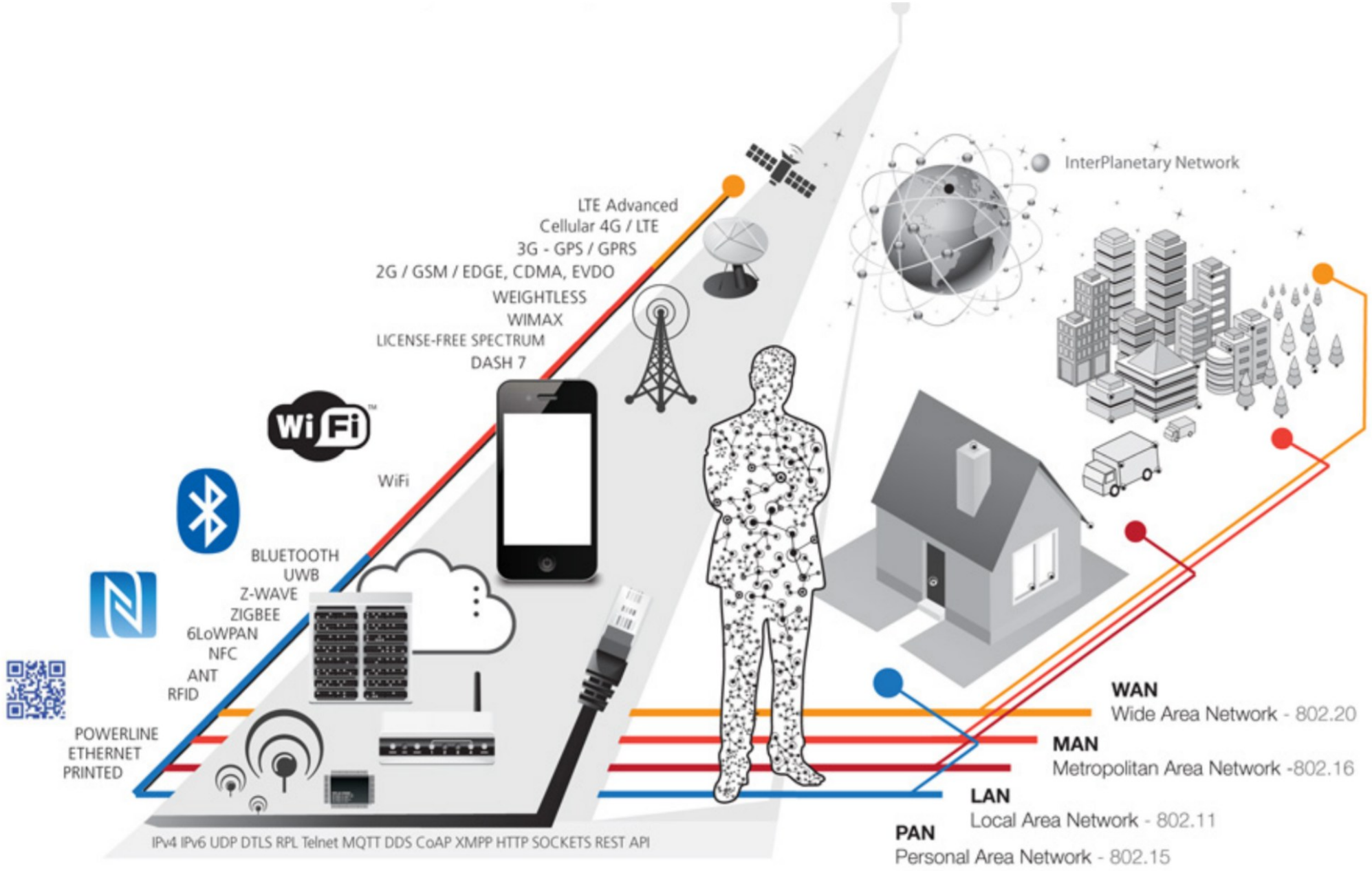
Network Size





- **WAN**: Wide Area Network
- **MAN**: Metropolitan Area Network
- **LAN**: Local Area Network
- **PAN**: Personal Area Network

Network Size

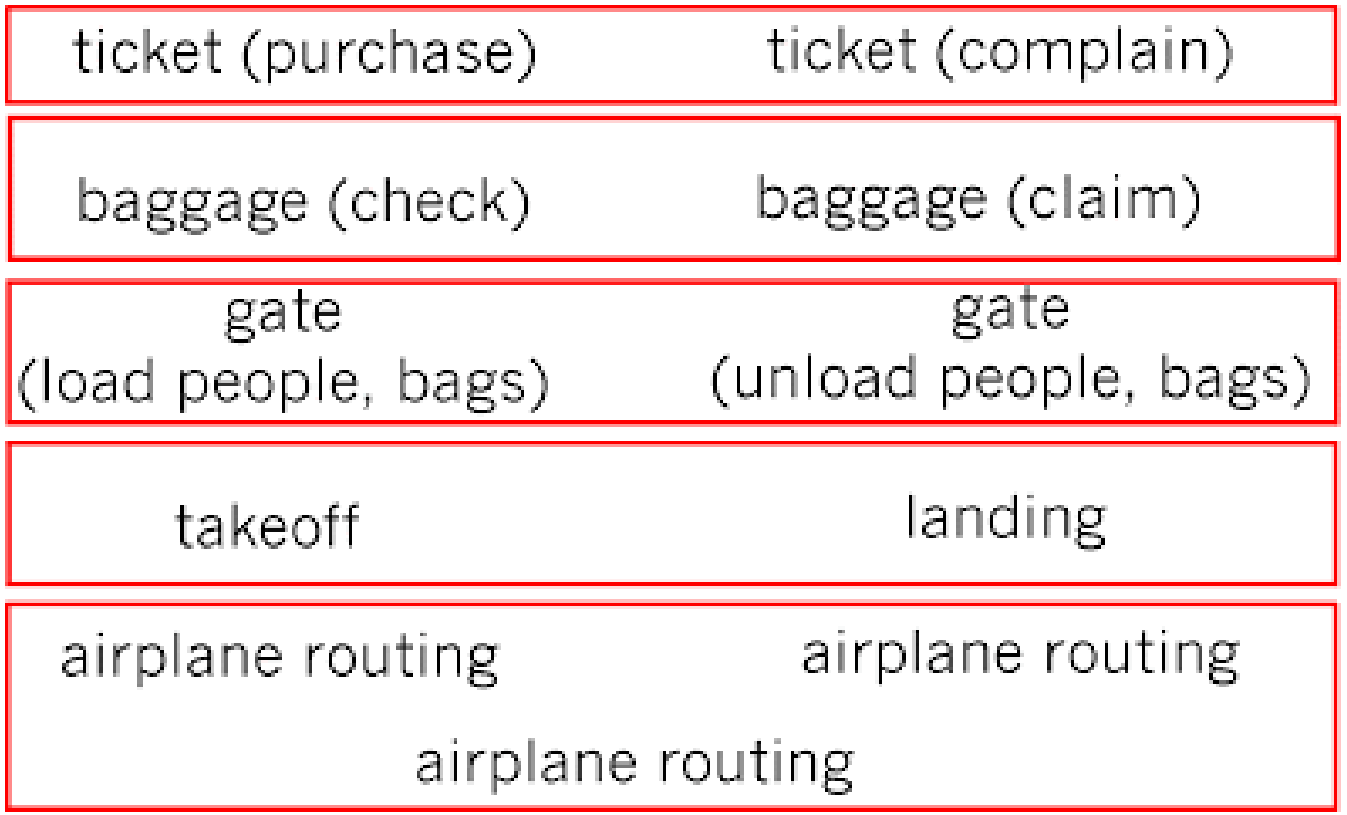


Layered Data Communication

- Data communication has become quite complex. Different operations are required to create an effective data communication system including
  - ✓ Signal to data conversion
  - ✓ Data unit encoding
  - ✓ Multiplexing and switching
  - ✓ Error management
  - ✓ Flow management
- In general, each operation should be separated from each other. An agent in each layer is not concerned with operations of agents in other layers. This is to provide modular independency for each layer.
- Layering allows hardware and software development of each module to be separated from each other. The development of one layer will not affect the development of another layer. This helps provide rapid advancement of technology.

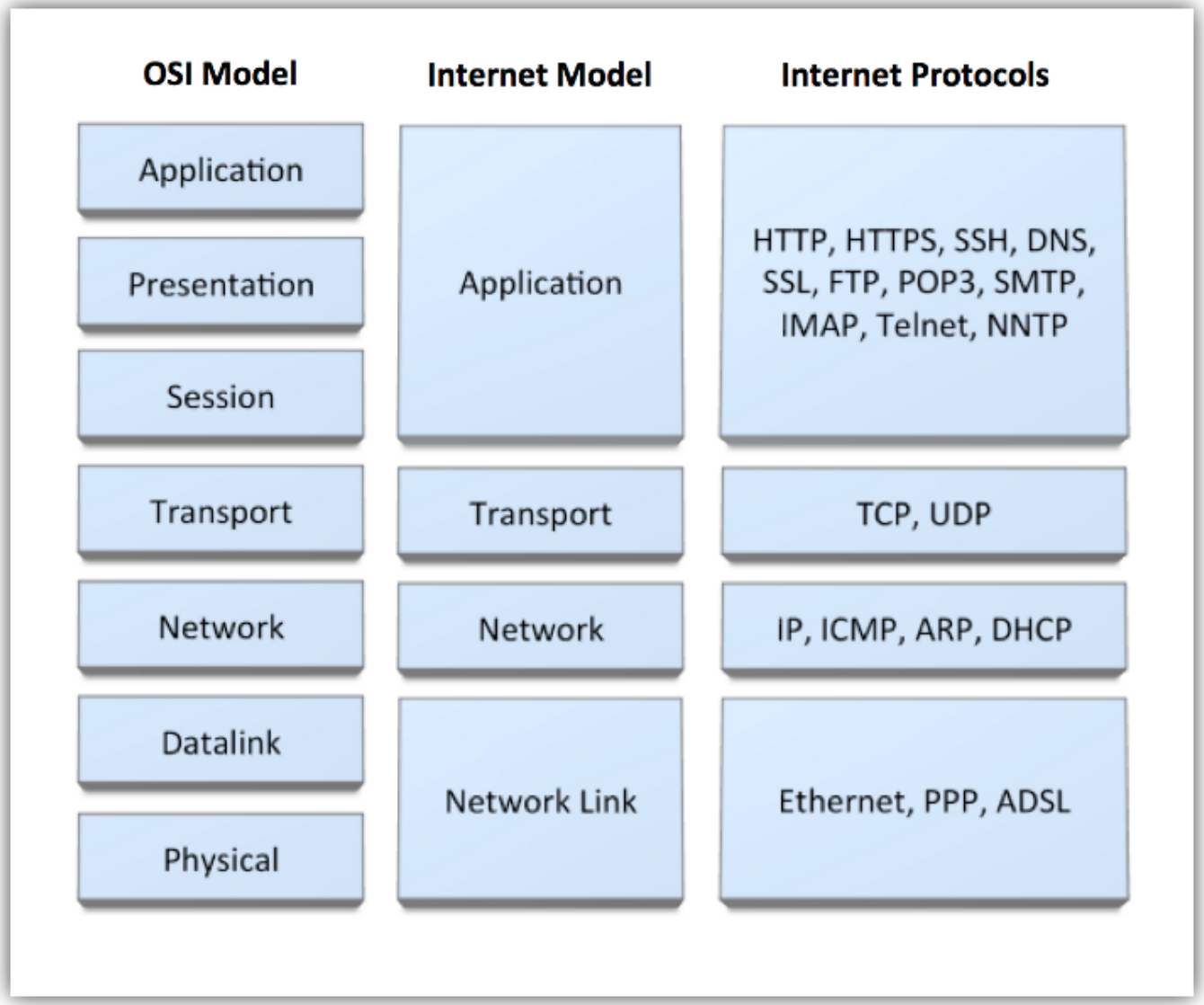
Layered Communication Analogy

- Airplane travel process is analyzed to show an example of layered operations.



Data Communication Models

All People Seems To Need Delicious Pizza



Data Units

Messages

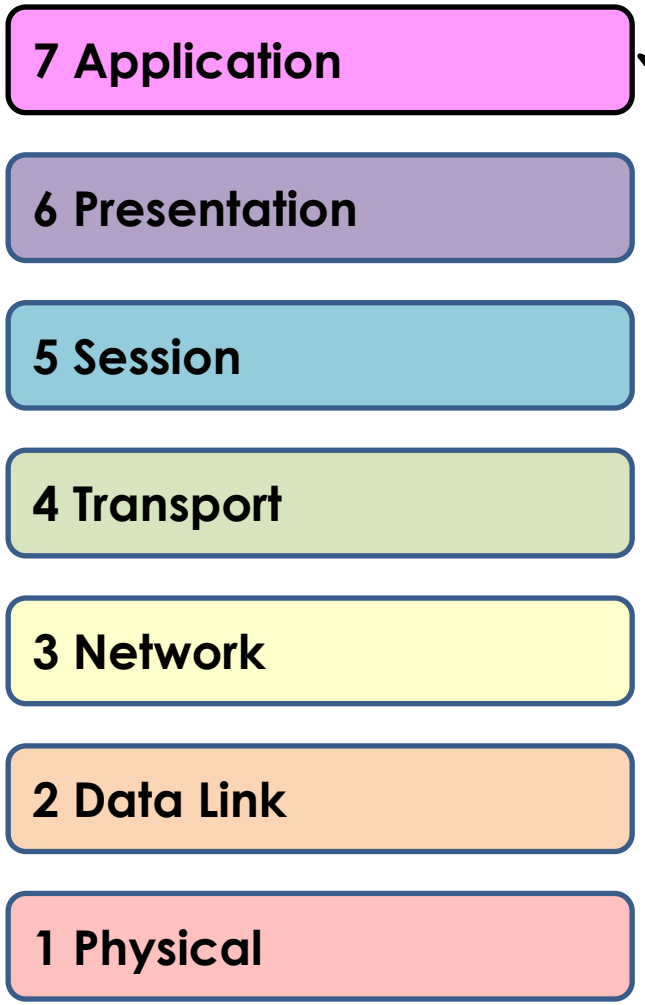
TCP segments / UDP datagram

Packets

Frames

Bits

OSI Layered Communication

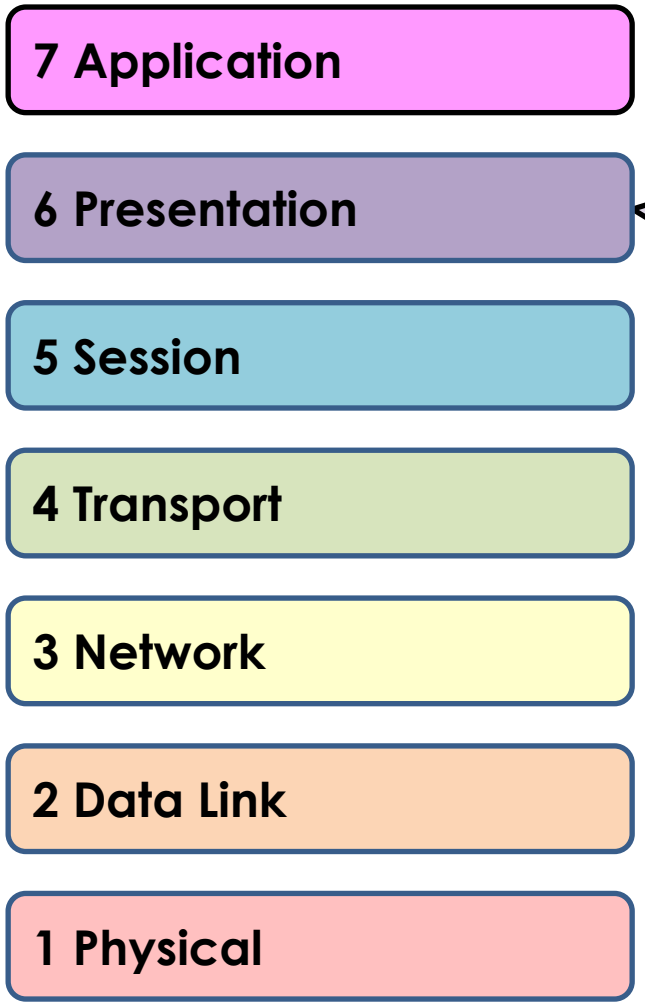


Application layer deals with networking applications. **Not only** the software itself **but also** the protocols that directly support the software applications. This layer carries user data or **messages**.

Features:

- E-mail (POP, IMAP, SMTP)
- Web (HTTP, HTTPS)
- Security (SSL)
- Remote login (Telnet, SSH)
- File Transfer (FTP)
- Network File System (NFS, WINS)
- Video Streaming (SIP, RTP, RTSP)

OSI Layered Communication

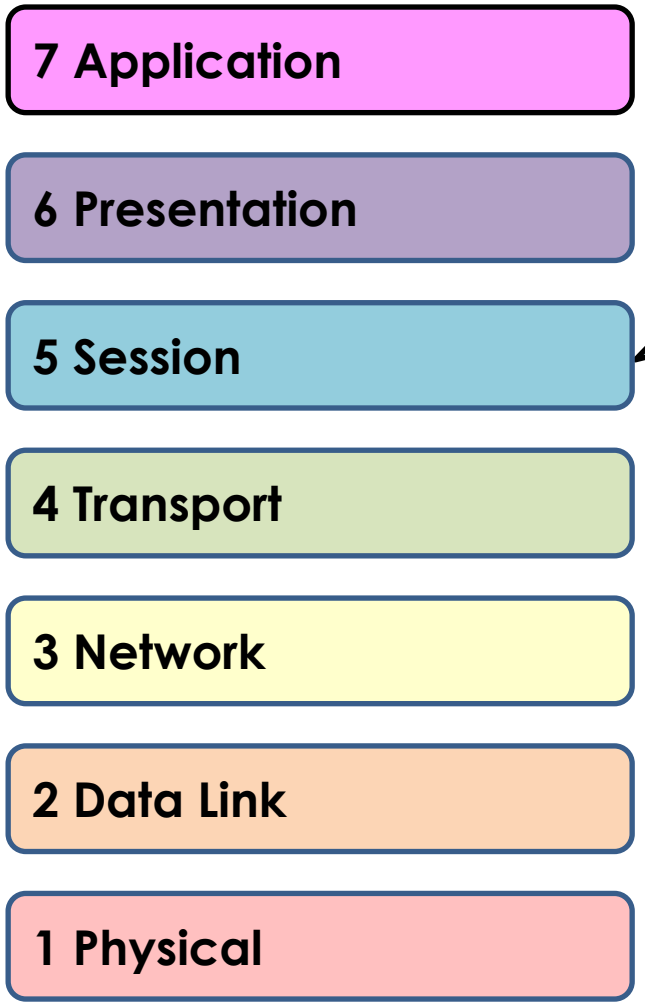


Presentation layer is responsible for presenting the data (message) in the required format agreed by both communicating parties. This layer carries **formatted data**.

Features:

- Character/code formatting (ASCII, UTF-8)
- Encryption (RSA, EC)
- Compression (JPEG, PNG, Zip

OSI Layered Communication

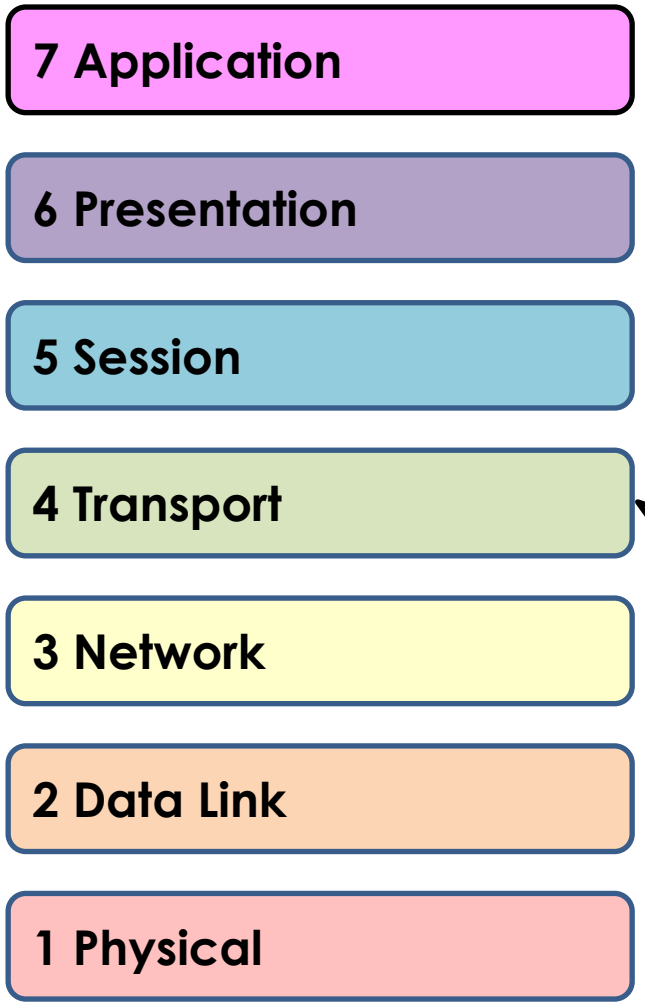


Session layer establishes, manages, and terminates sessions between two parties (multiplexing at the user level).

Features:

- Virtual Circuit/Best Effort
- Client-server sessions (HTTP, FTP, SSH)
- Login sessions
- Simplex, Half-Duplex, and Full-Duplex communication

OSI Layered Communication



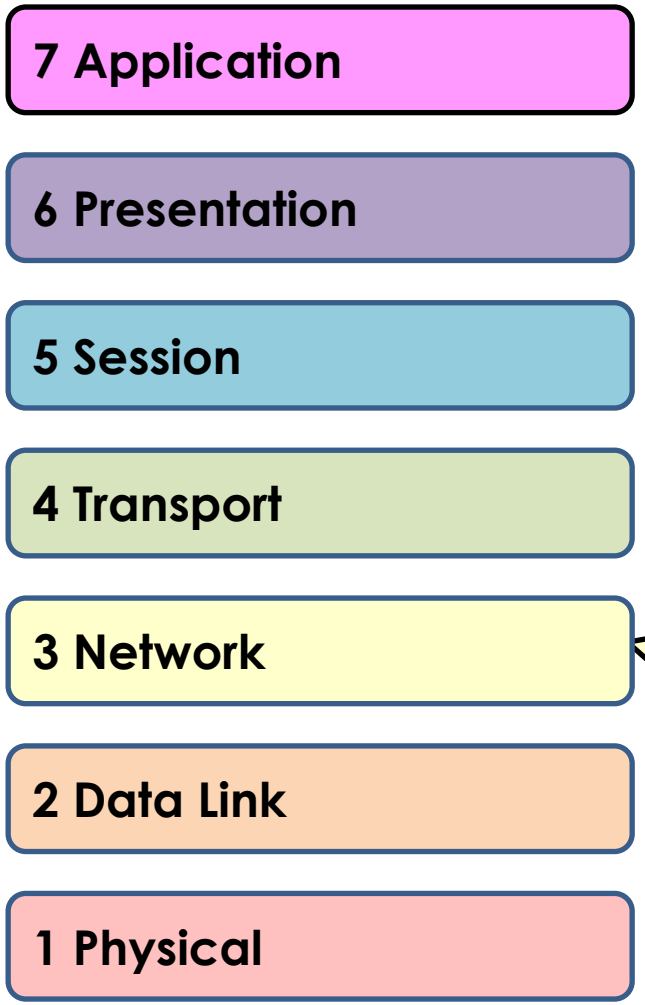
Transport layer (End-to-End)

- (sender) breaks up user data (message) into data **segments** and sends each segment to the receiver (either individually or group based on session layer)
- (receiver) reassembles data segments into original message and sends the message to the session layer.

Features:

- Segmenting
- Flow control
- Error control
- Session access/Multiplexing

OSI Layered Communication



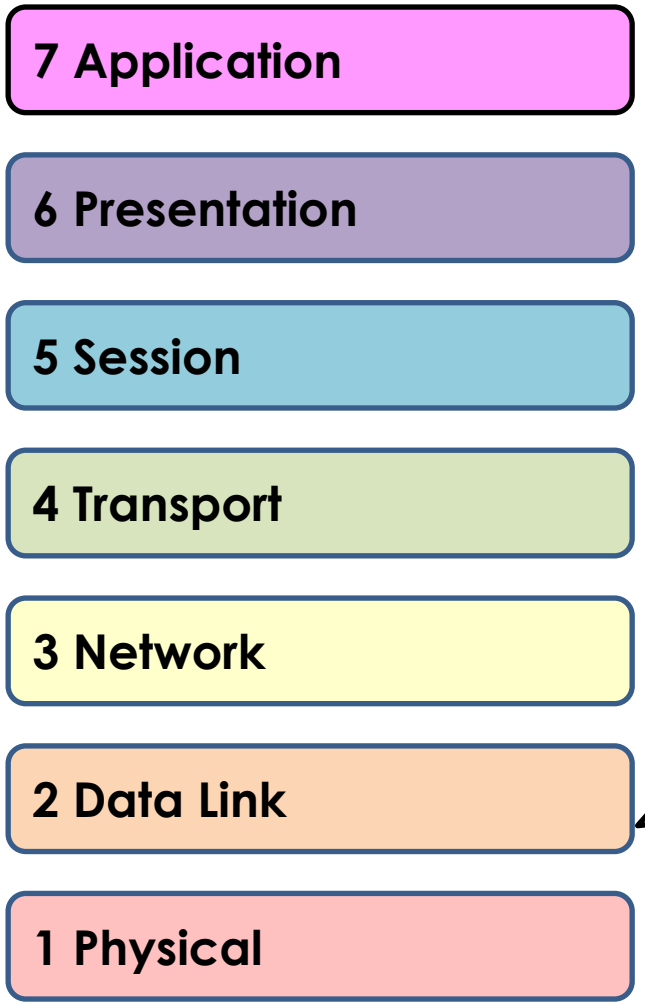
Network layer (Pathfinder)

- Provide end-to-end delivery with efficient and effective **route determination**.
- Data segments are added with postal service envelope (headers) to create data **packets** which will be delivered to the end point.
- Some segments may be broken into smaller packets.

Features:

- Shortest-path problem (Dijkstra, Bellman-Ford)
- Network information sharing (BGP)
- Network Management (ARP, ICMP, SNMP)
- Multi-label switching
- Addressing (IP, DHCP)

OSI Layered Communication



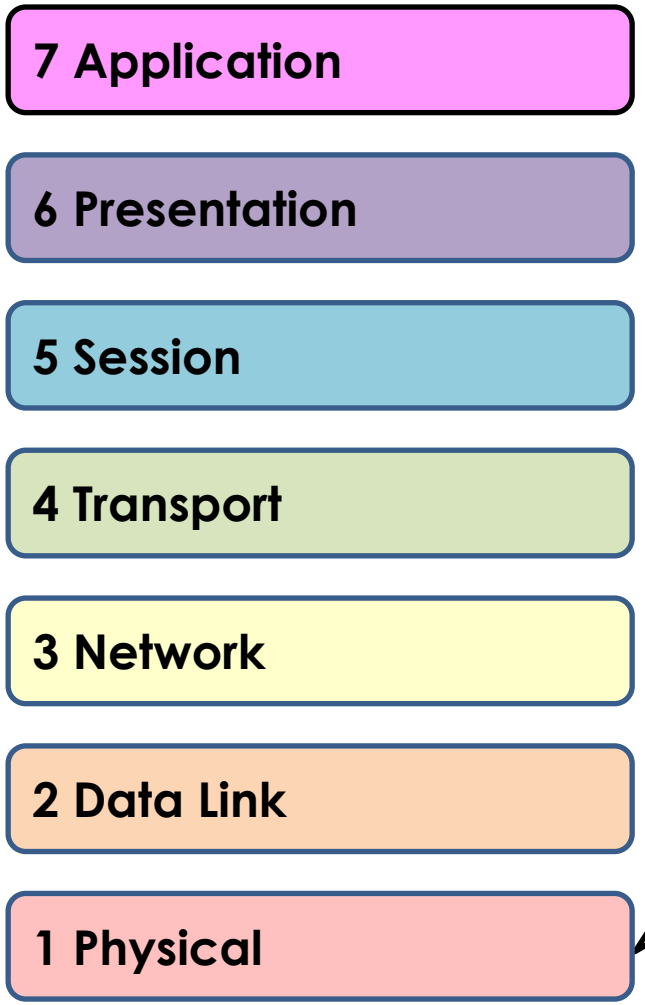
Data Link Layer (Node-to-Node)

- (sender) breaks up data packet into data **frames** and sends each frame to the direct receiver (either individually or group)
- (receiver) reassembles data frames into original packet and sends the packet to the network layer.

Features:

- Segmenting
- Flow control
- Error control
- Multiple Access/Switching/Multiplexing

OSI Layered Communication



Physical Layer (Cabling)

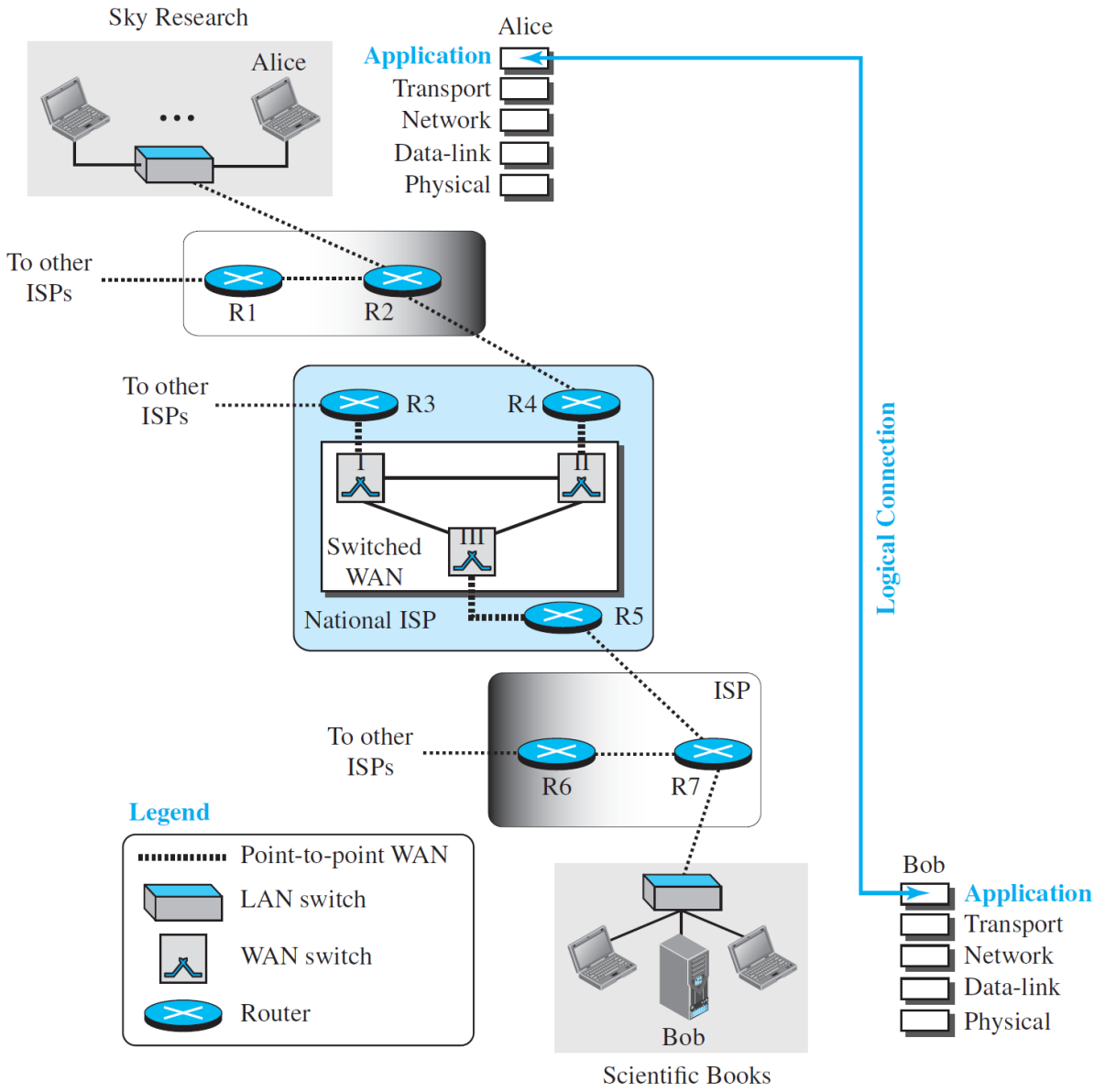
- Concerning the physical hardware of cables, interconnects, signal transmitter, signal receivers, and signal encoding.

Features:

- Analog and digital conversion.
- Cabling
- Signal manipulation and enhancement
- Network topology and architecture

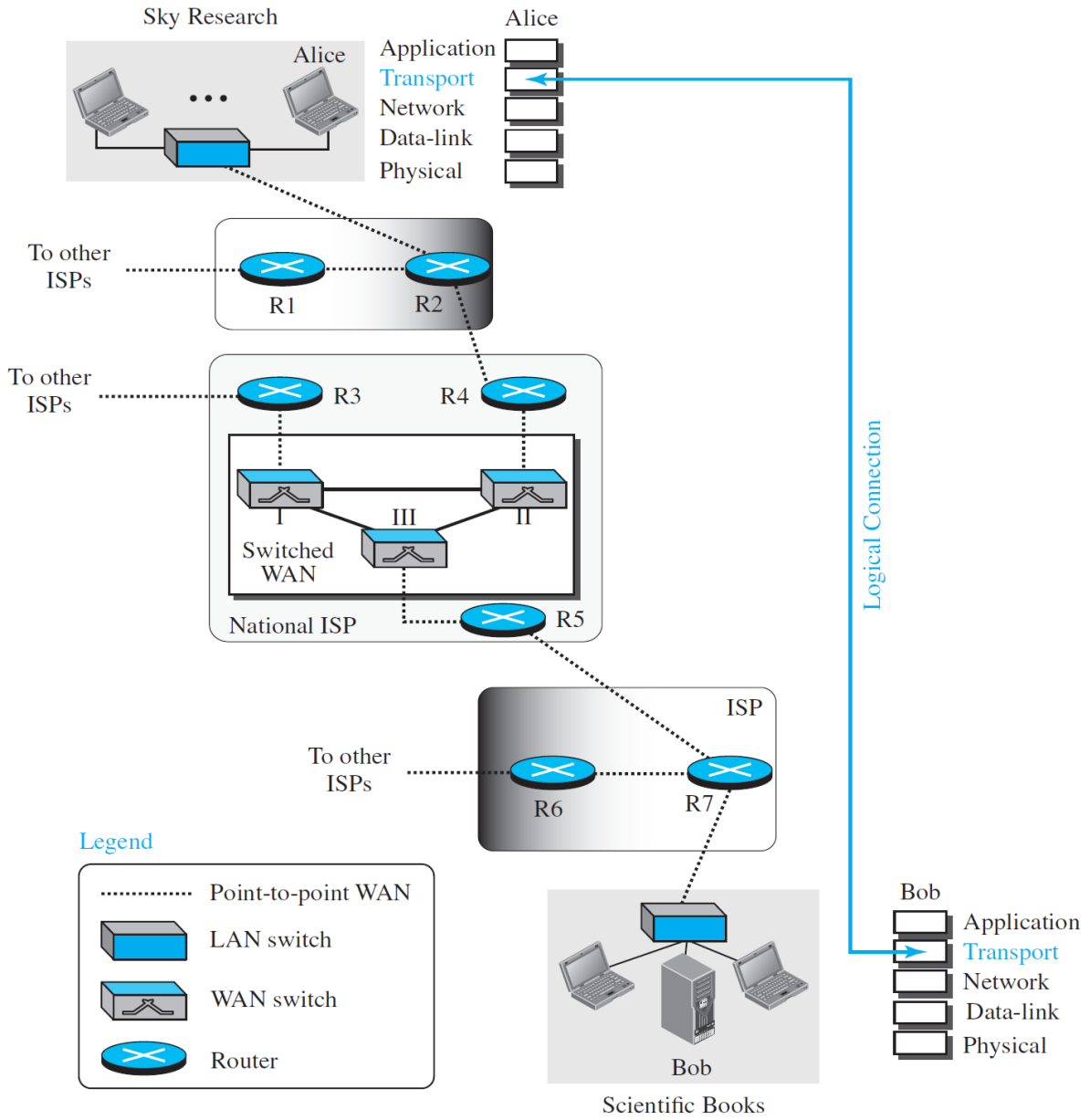
Communication at the Application Layer

Application layer communication (end-to-end)



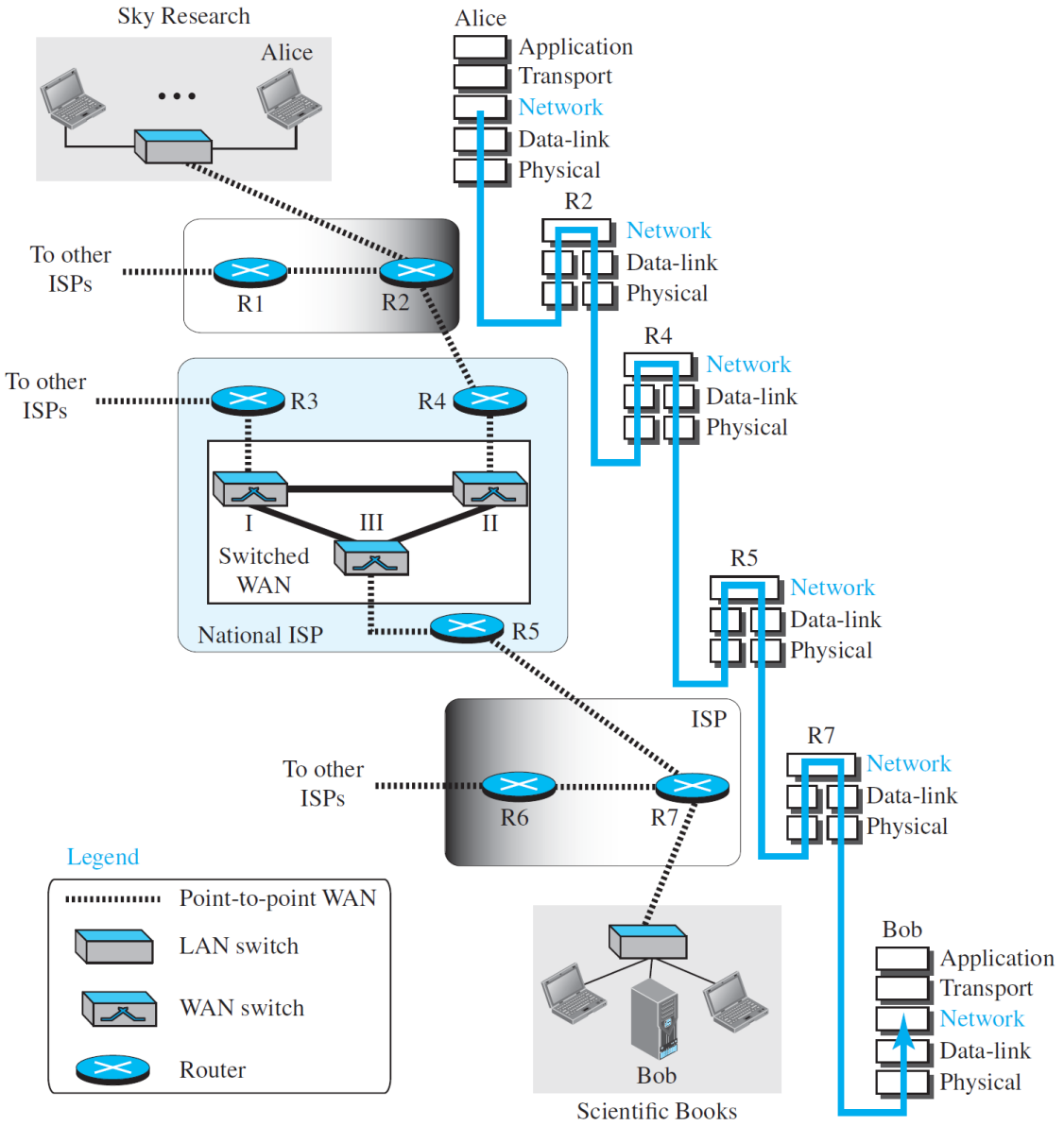
Communication at the Transport Layer

Transport layer communication (end-to-end or process-to-process)



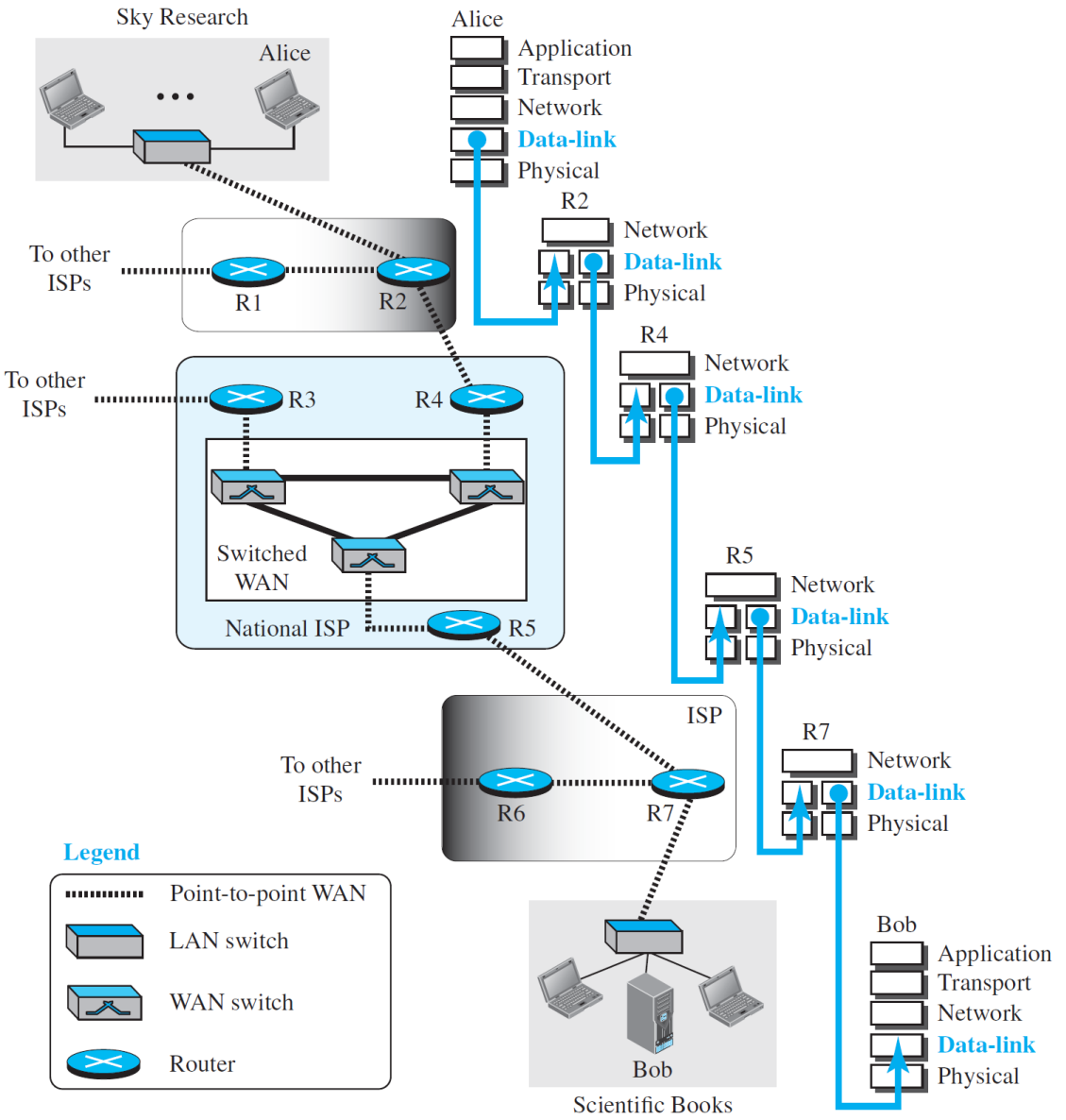
Communication at the Network Layer

Network layer communication (host-to-host)



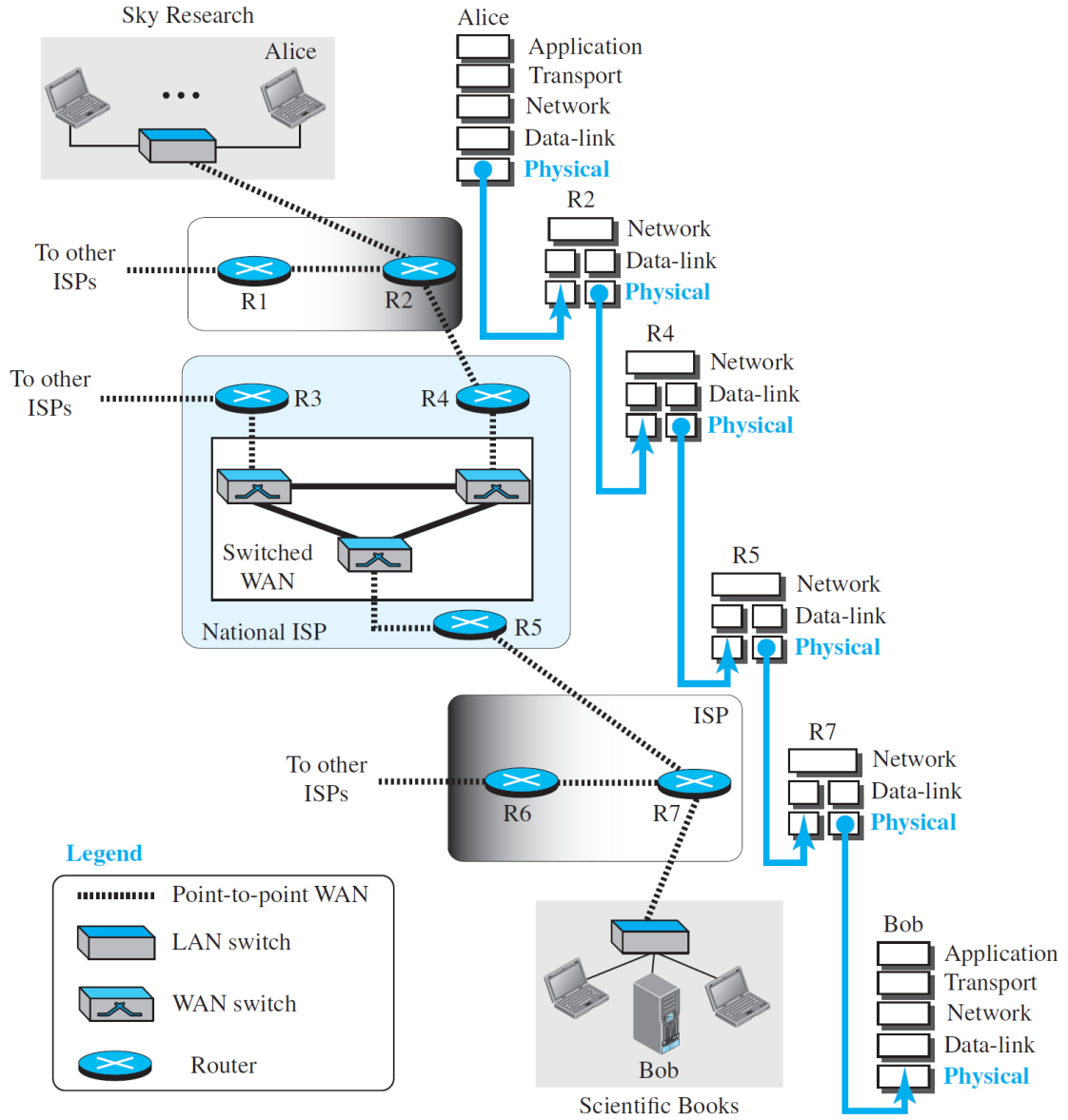
Communication at the Data Link Layer

Data link layer communication (node-to-node or hop-to-hop)



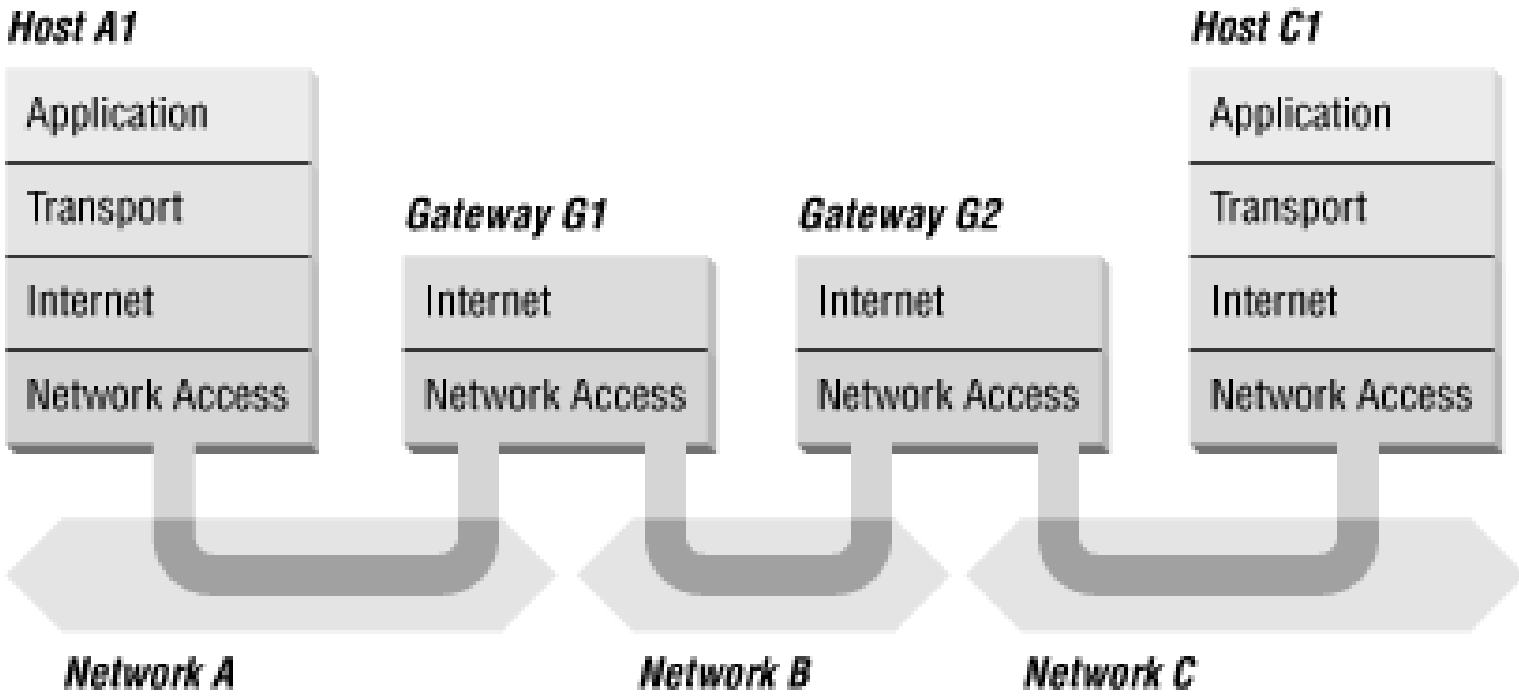
Communication at the Physical Layer

Physical layer communication (node-to-node)



Multi-hop Communication

- Not all layers are used by all devices.
- Some intermediate devices only use a few bottom layers.





Concepts for this Course

- How does the Internet work?
- How do we communicate using data?

This Photo by Unknown Author is licensed under CC BY-SA-NC

Exercise 1

1. _____ refers to the exchange of digital data between tow or more connected devices through a communication medium.

2. In data exchange, _____ refers to the agreed rules that determine how information is structured and interpreted, similar to grammar in a language.

3. _____ refers to the rules that define the structure, format, and arrangement of symbols in a message.

a. Packet Switching

b. Syntax

c. Semantics

d. Protocol

e. Internet

f. Delivery

g. Data Communication

h. Signal

Exercise 2

- Which OSI layer is responsible for **transmitting and receiving raw bits over the communication medium, including cables, connectors, voltage levels, signaling, and data rates**?

A. Application
B. Presentation
C. Session
D. Transport
E. Network
F. Data Link
G. Physical

Exercise 3

- Which OSI layer organizes data into frames and provides flow control, error control, and medium access between neighboring nodes?

A. Application
B. Presentation
C. Session
D. Transport
E. Network
F. Data Link
G. Physical

Exercise 4

- Which OSI layer is responsible for logical addressing, path determination, and packet delivery from a source host to a destination host across multiple networks?

A. Application
B. Presentation
C. Session
D. Transport
E. Network
F. Data Link
G. Physical

Exercise 5

- Which communication mode allows two-way communication between sender and receiver, but not at the same time?

A. Duplex
B. Simplex
C. Full-duplex
D. Half-duplex
E. Multiplex
F. Biplex

Exercise 6

- A security camera sends live video to a monitoring station. When the user sends a command to change the camera's direction or zoom, the video transmission pauses temporarily until the command is completed. Which communication mode is used?
  A. Simplex, because data travels in only one direction.
  B. Half-duplex, because both devices transmit at different times
  C. Full-duplex, because both devices transmit at the same time
  D. Multiplex, because several signals share one communication link
  E. Broadcast, because all connected devices can return information

Exercise 7

- Which of the following is NOT an appropriate example of simplex communication?
  A. A radio station transmitting programs to many listeners
  B. A television station broadcasting content to home viewers
  C. A computer sending display data to a basic monitor
  D. Two smartphones exchanging voice data during a call
  E. A public announcement system sending information to passengers

Exercise 8

- Which order correctly arranges network types from smallest to largest coverage?
  A. WAN → MAN → LAN → PAN
  B. PAN → LAN → MAN → WAN
  C. LAN → PAN → WAN → MAN
  D. MAN → WAN → PAN → LAN
  E. PAN → MAN → LAN → WAN

Exercise 9

- Which statement is **NOT** correct about communication protocols?
  - A. They define rules for sending and receiving data
  - B. They may specify message format and transmission timing
  - C. They help different devices communicate in an organized manner
  - D. They are required only when devices use wireless communication
  - E. They may include procedures for detecting communication errors

Exercise 10

- Which statement best explains an advantage of layered communication?
  - A. Hardware and software for one layer can be developed separately from other layers
  - B. All communication errors are completely eliminated by dividing operations into layers
  - C. Each layer can ignore the services provided by neighboring communication layers
  - D. Layering removes the need for flow control, encoding, and error management
  - E. Network communication becomes possible without standards or communication protocols

References

- https://en.wikipedia.org/wiki/History_of_telecommunication
- https://ethw.org/Morse_Code
- https://en.wikipedia.org/wiki/Morse_code
- Morse Code: https://www.youtube.com/watch?v=aniLPEyp9SA
- https://en.wikipedia.org/wiki/Data_communication
- https://en.wikipedia.org/wiki/History_of_the_Internet
- https://en.wikipedia.org/wiki/Computer_network
- OSI Layer Song
  - ✓ https://www.youtube.com/watch?v=7LazFWOfRm8
  - ✓ https://www.youtube.com/watch?v=kh7cZrzZnbA
  - ✓ https://www.youtube.com/watch?v=j7Bd9qwybAo
- Memorize OSI model
  - ✓ https://www.youtube.com/watch?v=KHMwhjQrCmo

